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NATIONAL INCOME

(National Income Meaning, Circular Flow of Income, National Income Aggregates and Measurement of National Income)

QUICK REVISION

NATIONAL INCOME

National income is defined as the monetary value of all final goods and services produced by the normal residents of the country whether operating within the domestic territory of the country or outside in a year. Points which should be noted in the definition of National income are:

1. Monetary Value: N.I. is always expressed in monetary terms. Since a variety of goods and services are produced in the country we need a common denominator to add them up. This money is used as a measuring rod to measure the value of all goods and services.

2. Final Goods Services: To measure N.I. we include value of final goods and services only because value of intermediate goods and services is already included in the value of final products. If we add value of intermediate goods in national income it will become double counting, i.e., adding value of a commodity in N.I. more than once. It gives an inflated N.I.

3. N.I. is a Flow: N.I. follows a flow concept because it is measured over a period time, i.e., 1 year, thus it is a flow of goods and services and not stock.

4. Current Output: N.I. includes the value of currently produced goods and services. Pure exchange transactions such as sale and purchase of second hand goods, securities and transfer payment are not included in N.I. because they do not contribute anything to the current year production. Sale and purchase of second hand goods are not included in N.I. but value of the services of commission agents involved in second hand sales and value of the services of brokers, i.e., brokerage is included in the N.I. Transfer payment is also not included in N.I. They are unilateral payments for which no productive services are rendered in current year, e.g., social security payments, donations, pensions, etc.

5. Normal Resident: N.I. is the value of the final goods and services produced by the normal residents of the country.Normal residents are those people who ordinarily reside in a country in which they live and whose economic interest lies in that country. They may or may not be the citizen of the country. It means normal residents include nationals as well as foreign nationals.

6. Domestic Territory: It refers to the geographical or political boundary of the country excluding foreign embassy and international institutions like U.N., W.H.O., offices located within the geographical territory and including the embassy of this country located outside its geographical territory.

CIRCULAR FLOW OF INCOME

Circular flow of income is defined as the flow of payments and receipts for goods and services and factor services among different sectors of an economy. There are four sectors in an economy, which are as follow:

1. Household Sectors

Features:
(i) They are owners of all factors of productions.
(ii) They receive income in the form of wages, rent, interest and profit. They also get some transfer payments from the Government.
(iii) They are the consumers of goods and services, i.e., they make consumption expenditure to business sectors. They also pay tax to the Government.
(iv) Savings by household sectors go to the capital market (Bank, stock market, etc.)
(v) Household sector exports its man-power to foreign sectors.
(vi) They make payments to foreign sector for whatever they import.

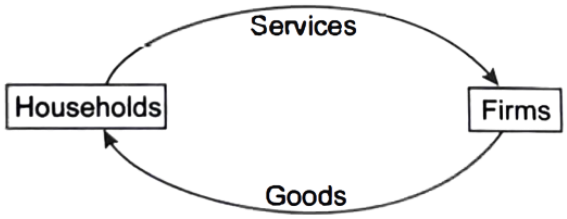
2. Firm/Producer/Business Sectors
Features:
(i) They hire factors of productions from the household sectors.
(ii) They produce and sell goods and services to the household sector and receive income from that sector.
(iii) They obtain subsidy from the Government.
(iv) They make factors payments to household sector and pay taxes to the Government. Savings by firms also go to the capital market.
(v) They take loan from capital market and invest in business sectors.
(vi) They earn export income.
(vii) They make payments to foreign sector to whatever they import.

3. Government Sectors

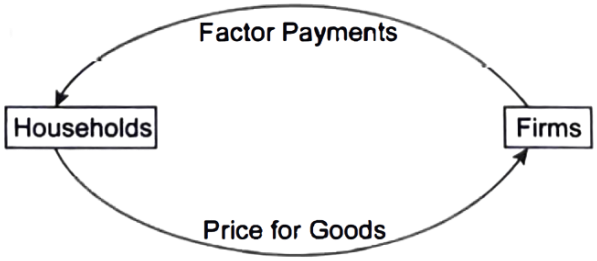
Features:

(i) Government sector receives direct taxes from household sector and indirect taxes from firms.
(ii) Government sector makes transfer payments to household sector and gives subsidies to business sector like food subsidy, fertiliser subsidy, etc.
(iii) Government savings also go to capital market.

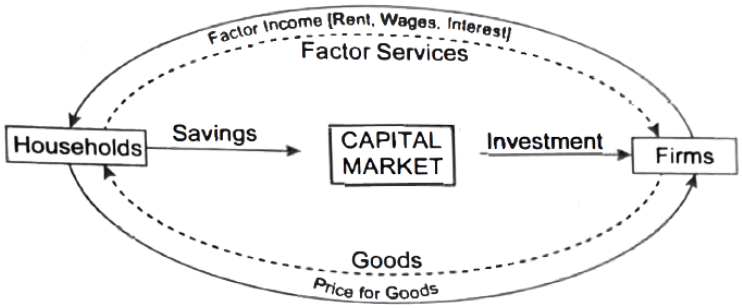
4. Foreign / External / Rest of the World Sector
Features:
(i) Rest of the world sector receives payments from the firms and household for their imports.
(ii) Rest of the world makes payments to the firm's imports.
(iii) External sector employs man-power from household sector and pays for them.
Types of Circular Flow of Income: Circular flow may be of two types: (i) Real flow, and (ii) Money flow.
(i) Real Flow: It shows the flow of goods and services through different sectors of economy.



(ii) Money Flow: Income flows as money flow shows flow of money across different sectors of economy.

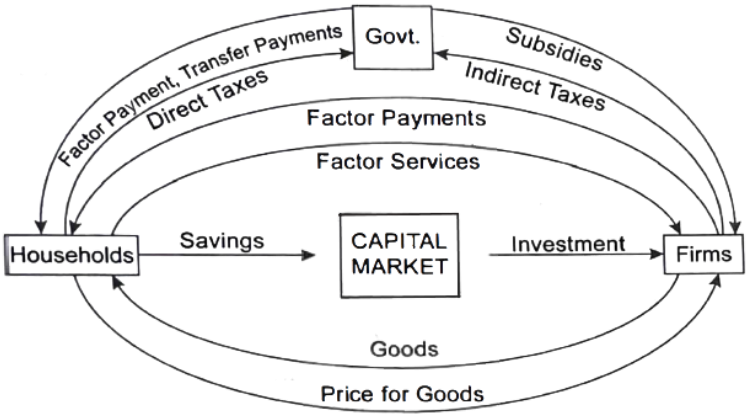


(a) Two-Sector Model of Circular Flow of Income: An economy consists of households and firms households, the suppliers of factor services to the firms, receive their rewards in form of wages, rent, interest and profits. They spend this money on the purchase of goods and services produced by the firms. Thus, income flows in a circular way between the households and the firms. Equilibrium in the two-sector model will take place when savings are equal to investments.



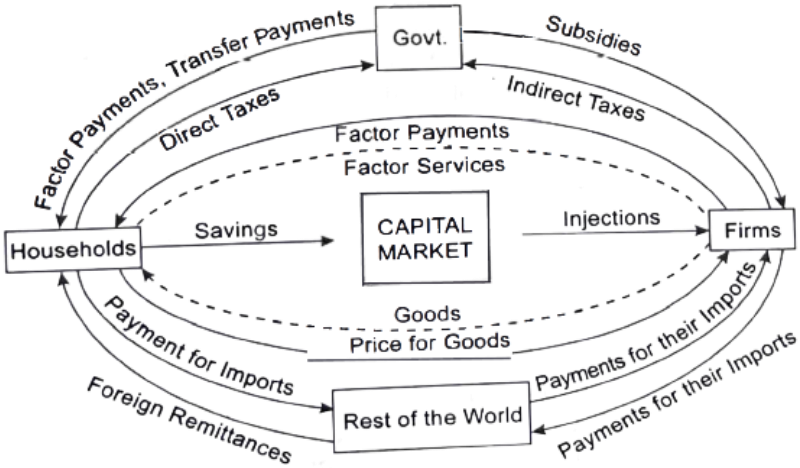
In this model, the condition for equilibrium is S = I or C + S = C + I. If S > I then the income flow decreases because S is known as withdrawal or leakage in circular flow. If S < I then the income flow increases because investment is known as injection in circular flow.

(b) Three-Sector Model of Circular Flow of Income: In this model three sectors, viz., households, firms and Government operate. Here the additional flows of income take place in form of taxes by the Government from household and firms, and flows of factor payments, transfer payments and subsidies from Government to household and firms.



In the above model leakages are savings and taxes. Injections are investment and Government expenditure on subsidies. If Savings + Taxes = Investments + Subsidies, then the flow will be in equilibrium. An economy with three sectors is known as a closed economy also.

(c) Four-Sector Model: A modern economy is a four-sector economy, where: (i) the households, (ii) the firms, (iii) the Government, and (iv) the rest of the world sectors exist. The circular flow of income goes on moving without interruption. In this model, the additional flows take place on account of exports and imports.



In the given model leakages are savings, taxes and imports. Injections are investment, Government expenditure and exports. Thus,

Savings + Taxes + M (Imports) = Investment + Govt. exp + X (Exports)

Then economy will be in equilibrium.

An economy with four sector is known as open economy also.

Leakages and Injections: Leakages are withdrawals like savings, taxes and imports. They reduce the flow of income. Injections are investments, subsidies and exports. They increase the flow of income.

Importance of Circular Flow Models: Circular flow models are important because they:

(i) show interdependence between sectors,

(ii) provide knowledge of structure of the economy,

(iii) facilitate the estimation of national income,

(iv) give information on important macro variables, and

(v) give information about injections and withdrawals.

NATIONAL INCOME AGGREGATES

1. GDP at Market Price: The market value of all final goods and services produced during a year within the domestic territory of the country. There are three important observations related to this:
(i) It is a gross product which means it includes depreciation in it.
(ii) It is a domestic product because it includes all goods and services which are produced in the domestic territory of the country. It does not matter whether the production is done by domestic or foreign companies.
(iii) Value of final goods and services, i.e., only the market value of final goods and services is taken into account.

2. GNP_{mp} at Market Price: GNP, the monetary value of all final goods and services produced in the domestic territory of the country during the year including net factor income abroad. Thus,

$$\mathbf{GNP_{mp} = GDP_{mp} + NFIA}$$

Or

It is a monetary value of all final goods and services produced by normal residents of a country whether operating within the domestic territory or abroad during a year.
The difference between GDP_{mp} and GNP_{mp} is NFIA. NFIA is the difference between income received from abroad for rendering factor services by the normal residents of the country to rest of the world and the income paid for factor service rendered by non-residents in the domestic territory of the country. Three components of NFIA are:
(i) Net compensation of employees from abroad, i.e., wages, in cash and kind both.
(ii) Net property and entrepreneurial income from abroad, i.e., rent, interest, profit, dividend, etc.
(iii) Net retained earnings of resident companies working in foreign countries.

NFIA can be both positive and negative. When NFIA is positive then GNP > GDP. If NFIA is negative then GNP < GDP and if NFIA is zero then GNP = GDP.

3. NDP at Market Price: NDP is market value of all final goods and services produced within the domestic territory of the country during the year excluding depreciation.

$$\mathbf{NDP_{mp} = GDP_{mp} - Depreciation}$$

Or

$$\mathbf{NDP_{mp} = GNP_{mp} - NFIA - Depreciation}$$

4. NNP at Market Price: NNP_{mp} is the market value of all final goods and services produced by the normal residents of the country during a year whether operating within the domestic territory or abroad by making allowances for depreciation.

$$\mathbf{NNP_{mp} = NDP_{mp} + NFIA}$$

$$\mathbf{= GNP_{mp} - Depreciation}$$

Or

$$\mathbf{= GNP_{mp} - Depreciation + NFIA}$$

5. GDP at Factor cost: It is the sum total of factors income (rent + interest + profit + wages) generated within the domestic territory of a country along with the consumption of fixed capital (Depreciation). Thus,

$$GDP_{fc} = GDP_{mp} - NIT$$

$$GDP_{rc} = GNP_{mp} - NFIA - NIT$$

6. GNP at Factor cost: It is the sum total of earnings received by normal reside nts of a country working as factors of production whether operating within the domestic country or outside in a year. Thus ,

$$GNP_{fc} = GDP_{fc} + NFIA$$

or $$GNP_{fc} == GNP_{mp} - NIT$$

or $$GNP_{fc} = NNP_{mp} - NIT + Depreciation$$

or $$GNP_{fc} = NNP_{fc} + Depreciation$$

7. NDP at Factor Cost : It is the sum total of factor incomes generated within the domestic territory of the country after deducting depreciation. Thus,

$$NDP_{fc} = GDP_{fc} - Depreciation$$

or $$NDP_{fc} = NDP_{mp} - NIT$$

or $$GNP_{fc} - Depreciation - NFIA = NDP_{fc}$$

or $$NNP_{fc} - NFIA = NDP_{fc}$$

It is known as domestic factors income. It includes:

(i) Compensation of Employees: It is the payment made by the producers in cash and kind both return of labour services. It includes wages and salaries in cash and kind and employer's contribution in social security schemes.

(ii) Operating Surplus: It is the income earned from property and entrepreneurship, e.g., rent, interest, profit, dividend, etc. Profit includes undistributed profit, dividend and corporation tax.

(iii) Mixed Income: Income earned by self employed people which is composite of labour income and property. Thus, NDP_{fc} **= Compensation of employees + Operating surplus + Mixed income**

NNP_{fc} or National Income

It is the sum total of factors income earned by normal residents of country during the year. It is also known as National Income.

$$NNP_{fc} = GNP_{fc} - Depreciation$$

or $$NNP_{fc} = NNP_{mp} - NIT$$

or $$NNP_{fc} = NDP_{fc} + NFIA$$

or $$NNP_{fc} = Dosmestic\ factor\ income + NFIA$$

Disposable Income Aggregates

Disposable income aggregates are derived from national income aggregates taking into account of certain types of transfer payments for which no productive services are rendered in the current year, e.g., old age pension, unemployment allowances, etc. (not included in N.I. as they do not contribute in current year.)

Transfer payments are of two types:
1. Current Transfer: It is one which is made out of the current income of the payer and adds to the current income of the receiver.
2. Capital Transfer: It is the one which is paid out of wealth of the payer and adds to the wealth of the receiver. In economics we are only concerned with the current transfer.

Disposable Income (DI)

Income inclusive of all net current transfer is called disposable income. Thus,

DI = Income + Net current transfer

Net current transfer = Current transfers received from - Current transfers paid to.

Net current transfer can be both positive and negative. If net current transfers are positive, disposable income will be higher than the income. On the other hand if net current transfers are negative, disposable income will be lower than the income.

National Disposable Income: It is defined as the sum total of national income at market price and net current transfers received from rest of the world. It is of two types:

(i) Gross National Disposable Income: GNDI = GNP_{mp} + Net current transfers from rest of the world.
(ii) Net National Disposable Income: NNDI = GNP_{mp} - Depreciation + Net current transfers from rest of the world.

PRIVATE DISPOSABLE INCOME AGGREGATES

1. Private income
2. Personal income
3. Personal disposable income

1. Private Income

It is the total factor income from all sources and current transform from the government and the rest of the world accruing to the private sector.
Private Income = NDP_{fc} - Income from the property and entrepreneurship accruing to Governmental administrative department - Savings of non-departmental enterprises + Net National debt interest + NFIA + Net current transfers from the Government + Net current transfers from rest of the world.

2. Personal Income (PI)

It is actual income received by persons from all sources in the form of factor income and current transfer payment during the year.

Pl = Private income – Undistributed profit - Corporate profit tax - Retained earnings of foreign companies.
It is useful in finding the purchasing power that is actually there in the hands of the people. PT is known as pre- tax income also.

3.Personal Disposable Income (PDI)

PDI is that part of Pl which is available to the individuals to be used the way they like.
PDI = PI - Personal taxes - Miscellaneous receipts of the Government administrative department (fines, fee).
PDI = Consumption + Saving.

Per Capita Income: PCI is the aggregate income of normal resident of the country in a particular year, thus

$$Per\ capita\ income(PCI) = \frac{Total\ national\ income}{Total\ population}$$

Per capita can be measured at current price as well as at constant prices. When it is measured at constant prices it is called real per capita income which is a very good indicator of economic growth. Formula to measure PCI is:

$$PCI = \frac{National\ income\ at\ constant\ prices}{Total\ population}$$

NI at Current Prices: When NI is measured at prevailing market price is known as Nl at current prices. It is called monitory or nominal national income also.

NI at Constant Price: When Nl measured at some base year price is called N at constant price. It is also known as real national income. Thus,

$$NI\ at\ constant\ price = \frac{NI\ at\ current\ price\ \times\ 100}{Price\ Index\ of\ current\ year}$$

Nominal GDP and Real DOP

Nominal UCP: Nominal GDP or GDP at current price refers to the production of goods and services valued at current prices.

Real GDP: GDP at constant prices refers to the production of goods and services valued at constant prices.

GOP Deflator: It measures the average level of prices of all the goods and services that make up GDP. It is used to eliminate the effect of price changes and to determine the real change in physical output.

$$GDP\ deflator = \frac{Nominal\ GDP}{Real\ GDP} \times 100$$

GOP and Economic Welfare: GDP is used as an indicator of economic welfare. However, it is not a satisfactory measure of economic welfare due to following reasons:

1. Composition of GDP
2. Changes in Prices
3. Non-monetary exchanges
4. Uneven distribution of GDP
5. Rate of population growth

Green SNP: It measures national income or output adjusted for the depletion of natural resources and degradation of the environment. It will help to attain a sustainable use of the natural environment and equitable distribution of developments. A larger number signifies greater sustainability.

MEASUREMENT OF NATIONAL INCOME

Methods of Measuring National Income: There are mainly three methods of measuring national income. These are: 1. Value added method, 2. Income method, and 3. Expenditure method.

1. Value Added Method (or Product Method): Under value added method, national income is estimated by taking and adding the money value of all final goods and services produced in an economy during a year. **Steps involved**: The following steps are involved while estimating national income by value added method:

(i). Identification and classification of productive enterprises.
(ii). Estimation of net value added.
(iii) Calculation of net factor income from abroad.
(iv) Estimation of national income.

Precautions:

(i) Sale and purchase of secondhand goods are not included.
(ii) Production for self-consumption should be included.
(iii) Services of owner-occupied houses should be included.
(iv) Own-account production of fixed capital should be included.

2. Income Method: Under this method, we take the sum total of factor incomes earned by the norm residents of a country during a year.
Steps Involved: The following steps are involved in the estimation of national income by using income method:
(i) Identification and classification of producing enterprises.
(ii) Classification of factor income.
(iii) Estimation of factor income.
(iv) Estimation of domestic factor income.
(v) Estimation of national income (i.e., NNP_{fc}).

Precautions:

(i) Transfer incomes should not be included.
(ii) Income from smuggling should be included but income through crimes as theft is not included.
(iii) Income from sale of secondhand goods should not be included.
(iv) Money received by way of sale of shares/bonds, etc., should not be included.
(v) Windfall gains do not form part of national income.
(vi) Corporation tax should not be recorded separately as it is a part of profit.
(vii) Death duties, wealth tax, gift tax, etc., are not included in the national income.
(viii) Imputed rent is a part of rental income.

3. Expenditure Method:
(i) For estimating national income according to expenditure method, we take final expenditures. Final expenditure can be estimated by adding the following:
(a) Final consumption expenditure,
(b) Final investment expenditure,
(c) Net exports.

(ii) The major components of the final consumption expenditure are as follows:
(a) Private final consumption expenditure,
(b) Government final consumption expenditure.
(iii) The major components of final investment expenditure are:
(a) Gross domestic fixed capital formation,
(b) Change in stocks.
(iv) By adding final consumption expenditure, final investment expenditure and change in stocks, we obtain GDP_{mp}

Precautions:
(i) Expenditure on secondhand goods is excluded.
(ii)Expenditure on shares, bonds, etc., should also be excluded.
(iii)Government expenditure on transfer payments should not be included.
(iv) Expenditure on intermediate goods should not be taken into account.

DIFFICULTIES IN THE MEASUREMENT OF NATIONAL INCOME

Conceptual Difficulties:
(i) Inclusion of services
(ii) Identifying intermediate goods
(iii) Identifying factor income
(iv) Services of housewives and other similar services
(v) Imputing unpaid services
(vi) Income of the foreign companies
(vi) Valuation of inventory changes
(viii) Estimation of depreciation

Practical Difficulties:
(i) Lack of occupational specialisation
(ii) Non-monetised sector
(iii) Inadequate information regarding income and output
(iv) Unreported illegal income
(v) Non-availability of reliable statistical data
(vi) Death of agencies
(vii) Collectors are semi illiterate and untrained
(viii) No accurate data regarding agricultural byproducts

1 MARK QUESTIONS

A. FILL IN THE BLANKS

NATIONAL INCOME

1. _____ variable is measured over a period of time.
2. Stock variables are measured at a particular _____.
3. Leakages refer to _____ of money from the circular flow.
4. Economy with international relations is known as _____.
5. Economy without international trade is known as _____.

6. National Income is a _____ concept.
7._____ transactions are not included in national income.
8. Exchange of money across different sectors of the economy is known as _____.
9. Income received after rendering factor services is known as _____.
10. Economic entities of an economy are _____ _____ _____ and _____.
11. Leakages of four sector model are _____ _____ and _____.
12. Injections of four sector model are _____ _____ and _____.
13. _____ = _____ at equilibrium level in a circular flow.
14. When S> I then flow _____.
15. When injections are more than leakages then flow _____.
16. _____ is a person who normally resides in the country for a period more than a year.
17._____ refers to a fall in the value of fixed assets due to normal wear and tear.
18. NIT refers to the difference between _____ and _____.
19. Value Added = _____ - _____.
20. Adding value of the same product in national income more than once is _____.
21. Double counting leads to _____.
22. Net Exports = _____ - _____.
23. Three methods of national income are _____ , _____ and _____.
24. Embassies of other countries are not included in the _____ of a country.
25._____ payments are not included in the national income of a country.

NATIONAL INCOME AGGREGATES

26. National Income - Domestic Income = _____.
27. Gross output _____ =Net Output
28. NIT = _____.
29. GDP_{mp}= NNP_{fc} + _____ - _____ + _____.
30. GDP_{fc} = GDP_{mp} _____ .
31. NNP_{mp}= GDP_{fc} _____ .
32 NFIA = GNP _____ .
33. Depreciation = GDP - _____ .
34. Domestic Factor Income = _____ + _____ + _____ .
35. NNP_{fc} = _____ + _____ .
36. GNP_{mp} = NDP_{fc} _____ + _____ + _____ .
37. Operating Surplus = _____ + _____ + _____ .
38. Leakages in a four sector economy = _____ .
39. Injections in an open economy = _____ .
40. Condition of equilibrium in a closed economy = _____ .

ANSWERS

1. Flow	2. Point of time	3. Withdrawal	4. Open economy
5. Closed economy	6. Flow	7. Pure	8. Money flow
9. Factor income	10.Household, Firms, Government, Rest of the world	11. Savings , Taxes, Imports	12. Investment, Subsidies, Exports
13. Leakages, Injections	14. Reduces	15. Increases	16. Normal residents
17. Normal residents	18. Indirect taxes, Subsidies	19. Value of output, Intermediate consumption	20. Double counting
21.Inflation	22. Imports, Exports	23. Product method, Income method, expenditure method	24. Domestic territory
25. Transfer payments	26. NFIA	27. Depreciation	28. Indirect Tax-Subsidy
29. Depreciation, NFIA, NIT	30. -NIT	31. _Depreciation + NFIA + NIT	32. -GDP
33. NDP	34. Compensation Of employees, Operating surplus, Mixed income	35, NDP NFIA	36. Depreciation,NFIA, NIT
37. Rent, Interest, Profit	38. Tax, Savings, Imports	39. Investment, Subsidies, Exports	40. Savings Taxes= Investments Subsidies

B. TRUE OR FALSE

1. National income includes both factor income and transfer income.
Ans. [False] Because it includes factor income only.

2. Wheat flour is always a consumer goods.
Ans. [False] If it is used by a restaurant it is an intermediate goods but true if purchased by a household.

3. Interest on national debt is included in national income.
Ans. [False] Because loans taken by the government are treated as loans for consumption purposes and it is treated as transfer payment.

4. Capital formation includes capital goods only.
Ans. [False] Because inventory investment is an important component of capital formation.

5. National income is a flow variable.
Ans. [True] Because national income is measured over a period of time.

6. Net exports are included in net factor income from abroad.
Ans. [False] Because exports are included in domestic products by expenditure method.

7. Services of a housewife are included in the national income.
Ans. [False] Because such services cannot be measured in money terms as they are performed out of love and affection.

8. A leakage is the amount of money which is withdrawn from its flow of money.
Ans. [True] For example, savings by households imply withdrawal from circular flow of money.

9. Real gross domestic product can be equal to nominal gross domestic product.
Ans. [True] If the price level in both the concerned years is the same.

10. Gross domestic product is always equal to gross national product.
Ans. [False] As it is only possible if NFIA is zero.

11. Gross domestic capital formation is always greater than gross fixed capital formation.
Ans. [False] Gross domestic capital formation can be less than gross fixed capital formation if change in stock is negative.

12. Capital formation is a flow
Ans. [True] Capital formation is measured over a period of time.

13. Traveling allowance paid by employers is a part of national income.
Ans. [False] Travelling allowances are paid by employees and then recovered from employer therefore it is not a part of national income.

14. Sale of secondhand car is not in national income.
Ans. [True] Because its original sale has already been included in national income of previous year.

15. Income from imputed rent of self owned houses is a part of national income.
Ans. [True] Because it is an estimated amount of rent payable to outsiders, if it is not owned property.

16. Change in stock is not a part of capital formation.
Ans. [False] Change in stock is a part of gross domestic capital formation as it is an addition to inventory.

17. Difference between value of exports and imports of goods and services is known as trade balance.
Ans. [False] Because the difference between the value of exports and imports of goods alone is known as trade balance.

18. Transfer income is included in national income.
Ans. [False] Because to earn transfer income, no productive activity is performed.

19. Embassies of other countries located in India will be included in the domestic territory of India.
Ans. [False] As they are included in the domestic territory of their own country.

20. NFIA is always positive.
Ans. [False] Because if factor income from abroad is less than factor income paid to abroad then NFIA will be negative.

QUESTIONS BASED ON REASONINGS ON NATIONAL INCOME

A. Giving reasons explain how the following are treated while estimating national income.

1. Payment of fees to a lawyer engaged by a firm.
2. Rent free house to an employee by an employer.
3. Purchases by foreign tourists.
4. Wheat grown by a farmer but used entirely for family's consumption.
5. Earning of the shareholders from the sale of shares.
6. Expenditure by the government on providing free education.
7. Remittance from non-resident Indians to their families in India.
8. Rent paid by the embassy of Japan in India to a resident Indian.
9. Profits earned by branches of foreign bank in India.
10. Salaries received by Indians residents working in Russian Embassy in India.
11. Profits earned by an Indian bank from its branches in abroad.
12. Entertainment tax received by the government.
13. Imputed rent of self-occupied house.
14. Interest received on debentures.
15. Financial help received by flood victims.
16. Capital gain on sale of a house.
17. Prize won in a lottery.
18. Interest on public debt.
19. Dividend received by an Indian from his investment in share of a foreign company
20. Money received by a family in India for relatives working abroad.
21. Interest received on loans given to a friend for purchasing a car.
22. Fees to a mechanic paid by a firm.
23. Interest paid by an individual on a car loan taken from a bank.
24. Expenditure on purchasing a car for use by a firm.
25. Taking care of aged parents.
26. Payment of corporate tax.
27. Expenditure on providing police service by the government.

B. Classify the following into gross fixed capital formation and change in stocks. Also give reasons.

1. Roads and bridges constructed by the government.
2. Increase in number of cows and buffaloes with a farmer.
3. New truck purchased by a transport company.
4. Construction of a new room by a consumer household.
5. Import of a secondhand rail engine from abroad.
6. Purchase of new houses by consumer households.

C. Questions based on reasonings on national income with answer.

1. How will you treat the following which estimating national income of India? Give reasons.

(i) Dividend received by an Indian from his investment in shares of a foreign company.
(ii) Money received by a family in India from relatives working abroad.
(iii) Interest received on loans given to a friend for purchasing a car.
(iv) Dividend received by a foreigner from investment in shares of an Indian company
(v) Profit earned by a branch of an Indian bank in Canada.

(vi) Scholarship given to Indian students studying in India by a foreign company.
(vii) Fees received from students.
(viii) Profits earned by branch of a foreign bank.
(ix) Interest paid by an individual on a loan taken to buy a car.
(x) Expenditure on machines for installation in a factory.
(xi) Profit earned by a branch of foreign bank in India.
(xii) Payment of salaries to its staff by an embassy located in New Delhi.
(xiii) Interest received by an Indian resident from firms abroad.
(xiv) Salaries received by Indians working in branches of foreign banks in India
(xv) Profits earned by an Indian bank from its branches aboards.
(xvi) Rent paid by embassy of Japan in India to an Indian resident.
(xvii) Imputed rent of self occupied house.
(xviii) Interest received on debentures.
(xix) Financial help received for flood victims.

Ans. (i) It is factor income from abroad so will be included in national income.
(ii) It is transfer receipts, so it is not included in national income.
(iii) Not included in national income, because it is a non-factor receipt as loan is not used for production for consumption
(iv) Included as it is a factor income to abroad.
(v) It is a part of NFIA and will be included in national income.
(vi) It is transfer receipts, so it is not included in national income.
(vii) It is included in national income because it is a part of the private final consumption expenditure of the house hold.
(viii) Included in national income because it is part of domestic factor income of India.
(ix) Not included because it is a non-factor income as loan is not used for production but for consumption.
(x) Included because it results in flow of income throught productive activities
(xi) Included, because it is a part of domestic product of India.
(xii) Not included because it is not a part of domestic product of India.
(xiii) Included as it is the part of NFIA.
(xiv) Included because it is earned in domestic territory of India.
(xv) Included because it is a part of NFIA
(xvi) Included as it is paid to an Indian resident outside the domestic territory of a country. It will be included in NFIA
(xvii) Included as a part of rent as it is payment to self for housing services.
(xviii) Included because it is a factor earning.
(xix) Not included as it is a transfer payment.

2. How will you treat the following which estimating domestic factor income of India? Give reasons

(i) Remittances from non-resident Indian to their families in India
(ii) Rent paid by the embassy of Japan in India to a resident Indian.
(iii) Profit earned by branches of foreign bank in India.
(iv) Payment of salaries to its staff by embassy located in India.
(v) Interest received by an Indian resident from firms abroad.

Ans. (i) Not included as it is a transfer payment.
(ii) Not included because Japanese embassy in India does not fall with in the domestic territory of India.
(iii) Included because it falls with in the domestic territory of India.
(iv) Not included as an embassy located in India is not fall with in the domestic territory of India.
(v) Not included in domestic product but it is the part of NFLA

3. Are the following part of a country's net domestic product at market price? Explain :
(i) Net indirect tax
(ii) Net export
(iii) NFIA
(iv) Consumption of fixed capital
Ans. (i) Yes, because market price = Factor cost + Net indirect tax.
(ii) Yes, because NDPMP includes net exports.
(iii) No, because domestic means it excludes NFIA.
(iv) No, net means consumption of fixed capital is excluded.

4. Will the following be included in Gross Domestic Product/Domestic Factor Income of India? Give reasons for each answer.
(i) Old age pension given by govt.
(ii) Factor income from abroad.
(iii) Salaries to Indian residents working in American embassy in India.
(iv) Compensation of employees given to residents of China working in Indian embassy in China.
(v) Profit earned by a company in India, which is owned by a non-resident.
(vi) Profit earned by an Indian company from its branch in Singapore.
Ans. (i) No, because pension is paid on account of old age of a pensioner and not for his rendering productive services.
(ii) No, because factor income is earned not within the domestic territory of a country but from abroad.
(iii) No, because American embassy is not a part of domestic territory of India.
(iv) Yes, because Indian embassy in China is a part of domestic territory of India.
(v) Yes, because the company within India's domestic territory earns profit.
(vi) No, because the branch is located outside the domestic territory of India.
5. Are the following included in the estimation of National Income of India? Give reasons for each answer
(i)Profit earned by a foreign company bank in India.
(ii) Money received from sale of shares.
(iii) Salary paid to Americans working in Indian embassy in America
(iv) Salary paid to Indians working in Indian embassy in America.
(v) Scholarship received by a student.
(vi) Remittances from aboard.
Ans. (i) No, as it is a factor income paid abroad (it is earned by non-residents).
(ii) No, it is only a transfer of paper claims.
(iii) No, this factor income belongs to non-residents.
(iv) Yes, as it is a factor income paid to normal resident of India.
(v) No, it is only a transfer payment.
(vi) No, it is only a transfer payment. No commodity is sent or services rendered return for this.

6. Will the following be included National Income? Give reasons for each answer.
(i) Services of owner occupied houses.
(ii) Purchase of new shares of a domestic firm.
(iii)Purchase of second-hand machine from a domestic firm.
(iv) Consultancy fee paid to a foreign expert.
(v) Commission paid to agent for the sale and purchase of shares.
(vi) Dividend received on shares.
Ans. (i) Yes, imputed rent of owner occupied houses will be included in Ni
(ii) No, because it is a financial transaction which does not help directly in production.
(iii) No, because it is not related with current flow of goods and services.
(iv) No, as it is a factor income paid abroad (it is earned by non-residents).
(v) Yes, it is included in NI since it is paid for rendering productive services.
(vi) Yes, dividends are a part of corporate profit and therefore, include in NI

7. Will the following be included National Income? Give reasons for each answer.
(i) Free Medical facility to employees by the employer. |
(ii) Money received from sale of old house.
(iii) Government expenditure on street lighting.
(iv) Interest received by a household from a commercial bank.
(v) Receipts from sale of land.
(vi) Interest on public debt.
Ans. (i) Yes, as it is a supplementary income paid in kind and hence a part of compensation of employees.
(ii) No, as it has already been taken into account when the house was constructed.
(iii) Yes, it is a part of Government final consumption expenditure and it adds to flow of services.
(iv) Yes, as it is payment for use of capital.
(v) No, as it does not add to flow of goods and services.\
(vi) It should not be included in Ni because public debt is a loan taken on to meet consumption expenditure by the government.

8. Are the following included in the estimation of National Income a country? Give reasons.
(i) Services rendered by family members to each other.
(ii) Wheat grown by a farmer but used entirely for family's consumption.
(iii) Expenditure government on providing free education.
(iv) Payment of fees to a lawyer engaged by a firm.
(v) Man of the match award to a player of the Indian cricket team.
(vi) Payment of the match fee to players of Indian cricket team.
Ans. (i) Services rendered by family members to each other should not be included in Ni because these are not rendered for the purpose of earning income.
(ii) Imputed value of self-consumed wheat grown by a farmer must be included in NI, because it adds in the flow of goods
(iii) It should be included in Ni because the government expenditure on the free services is considered as a part of government final consumption expenditure.
(iv) Yes, as it is factor income against the service of lawyer.
(v) It should not be included in NI because it is a windfall gain and it does not add in the flow of goods and services.
(vi) It should be included in NI of India because they render productive services as professionals.

9. Are the following included in the estimation of National Income a country? Give reasons.
(i) Unemployment allowance under NREGA.
(ii) Indirect tax (Sale tax/excise duty).
(iii) Salary received by the workers under NREGA.
(iv) Income tax.
(v) Corporation tax.
(vi) Travelling expenses paid to salesman by the employer.

Ans. (i) It is transfer payment received by those persons who are not employed; therefore it should not be included in NL
(ii) It is not included in NI because it does not add in the flow of goods and services.
(iii) It is included in NI because it is a factor income.
(iv) It is a part of compensation of an employee (income). While calculating NI by income method, compensation of employees is to be included while doing so, income tax to be paid by them should not be included separately.
(v) It is a part of profit of corporate sector. While calculating NI by income method, profit is to be included while doing so, corporation tax should not be included separately.
(iv) Travel expenses incurred by exployees for business purpose which are reimbursed by the employers are excluded because these are a part of intermediate consumption of the employers.

C. MULTIPLE CHOICE QUESTIONS

NATIONAL INCOME

1. Circular flow of income, flows between:
(a) production, income and expenditure (b) consumption and production
(c) firm and industry (d) different components of society
2. Which of the following is included in real flow?
(a) Flow of goods (b) Flow of services (c) Both (a) and (b) (d) None of these
3. What is included in circular flow?
(a) Real flow (b) Money flow (c) Both (a) and (b) (d) None of these
4. Which of following is not included in the structure of an economy?
(a) Household sector (b) Firm sector (c) Poor sector (d) Government sector
5. Which of the following is a closed economy?
(a) Economy without Government interference (b) Economy without international trade
(c) Free economy (d) All of these
6. Which is main components of capital market?
(a) Consumption and investment (b) Income and saving
(c) Saving and investment (d) None of these
7. What is main function of household sector?
(a) Aggregate consumption (b) Production
(c) Saving and investment (d) None of these
8. Which of following statements is not true?
(a) Capital sector is a component of macro economics (b) Income is a circulate flow
(c) Rest of world sector is found in closed economy (d) All of these

9. Which of following is not included in real flow?
(a) Goods (b) Services (c) Money (d) All of these
10. In three-sector circular flow of income, which of following is not included?
(a) Household sector (b) Firm sector (c) Government sector (d) Rest of world sector

ANSWERS

1. (a)	2. (c)	3. (c)	4. (c)	5. (b)	6. (c)	7. (a)	8. (c)	9. (c)	10. (d)

MEASUREMENT OF NATIONAL INCOME

1. GNP_{mp} = ?
(a) GNP_{mp}-Depreciation (b) GNP_{mp} + Net factor income from abroad
(c) GNP_{mp}+ Subsidy (d) None of these
2. Which of the following is not included in primary sector?
(a) Farming (b) Mining (c) Manufacturing (d) All of these
3. What is an alternative name of secondary sector?
(a) Production, income and expenditure (b) Consumption and production
(c) Firm and industry (d) Different components of society
4. Which of the following is not included in secondary sector?
(a) Small industry (b) Large industry
(c) Banking and insurance (d) All of these
5. What do we call the service sector?
(a) Primary (b) Secondary
(c) Tertiary (d) None of these
6. Which of the following is not included in tertiary sector?
(a) Banking and insurance (b) Transportation
(c) Small scale industries (d) Telecom industry
7. In total expenditure method which of following sector's expenditure is included?
(a) Household sector (b) Government sector
(c) Firm sector (d) All of these
8. What is not included in gross domestic capital formation?
(a) Investment in machinery (b) Investment in stocks
(c) Investment in shares (d) Investment in valuable
9. Which of the following expenditure on goods is not included in national income?
(a) Old goods (b) Shares (c) Smuggled goods (d) All of these
10. Exports are:
(a) expenditure (b) income (c) transfer earning (d) factor income
11. Which of the following is not included in national income?
(a) Factor income (b) transfer income
(c) Operating surplus (d) Net factor income from abroad
12. Which of the following is a transfer earning?
(a) Earned income (b) Factor income (c) unearned income (d) All of these
13. Which of the following is included in gross national income?
(a) Mixed income (b) Depreciation cost (c) Interest and Rent (d) All of these

14. Which of the following is not included in net value added at market price?
(a) Net indirect taxes (b) Total production (c) Depreciation cost (d) None of these
15. Which of the following is not included in net value added at market price?
(a) Depreciation cost (b) Net indirect tax
(c) Transfer earning (d) None of these
16. Which of the following is not included in national expenditure?
(a) Purchase of machinery (b) X-M
(c) Help to accident victims (d) None of these
17. Which of the following is not a domestic factor income?
(a) Net exports (b) Wages and salaries
(c) Salary of Indian workers working in Russian embassy (d) None of these
18. Net value added at factor cost is equal to:
(a) national income (b) net domestic product
(c) gross domestic product (d) none of these
19. How double counting can be avoided by using?
(a) Final goods method (b) Value added method (c) Any of these (d) none of these
20. Expenditure on road construction is which type of expenditure?
(a) Household final expenditure Govt final expenditure (b) X - M
(c) Transfer payments (d) Govt. final expenditure
21. Which of the following is not included in national income?
(a) Compensation of employee (b) Income from sale of shares
(c) Operating surplus (d) Factors income
22. Which of the following is true?
(a) Value added method should be adopted to avoid double counting (b) NNPfc -GNPfc - Depreciation
(c) Transfer income is not included in national income (d) All of the above
23. Which of the following is not a part of domestic income?
(a) Transfer earning (b) Net factor Income from abroad
(c) Sale of old goods (d) All of these
24. Which of the following is not a part of compensation of employees?
(a) Salary and wages (b) Commission and brokerages
(c) Bonus (d) Travellers allowance
25. Which of the following is not included in operating surplus?
(a) Profit (b) Royalty
(c) Rent (d) None of these
26. Which of the following is not included in net value added at factor cost?
(a) Factor's income (b) Sale of services
(c) Net indirect taxes (d) None of these
27. Which of the following is included in national income?
(a) Transfer income (b) Income from sale of old car
(c) Imputed rent (d) Income from black marketing
28. For a calculation of national income, income from which of following sector is included?
(a) Primary (b) Secondary
(c) Tertiary (d) All of these

29. Which of the following is true?

(a) GNP GDP + Depreciation
(b) NNP = GNP + Depreciation
(c) NNP = GNP – Depreciation
(d) GNP NNP - Depreciation

ANSWERS

1. (b)	2. (c)	3. (c)	4. (c)	5. (c)	6. (c)	7. (d)	8. (c)	9. (d)	10.(a)
11.(b)	12.(c)	13.(d)	14.(c)	15.(c)	16.(c)	17.(c)	18.(b)	19.(c)	20.(d)
21.(b)	22.(d)	23.(d)	24.(d)	25.(d)	26.(c)	27(c)	28.(d)	29.(c)	

NATIONAL INCOME AND RELATED AGGREGATES

1. Which of the following included depreciation?

(a) GDP_{mp}
(b) NNP_{mp}
(c) NNP_{fc}
(d) None of these

2. Which of the following is not included in primary sector?

(a) Factor income (b) Rent (c) Operating surplus (d) Transfer income

3. Which of the following is not a part of national income?

(a) Depreciation
(b) Net factor income from abroad
(c) Operating surplus
(d) Mixed Income

4. Which of the following includes net indirect tax?

(a) GNP_{fc}
(b) NNP_{fc}
(c) NNP (d) None of these
(d) None of these

5. Which of the following does not include net factor income from abroad?

(a) GNP_{fc}
(b) NNP_{fc}
(c) NDP_{fc}
(d) Private income

6. Which of the following is known as national income?

(a) NNP_{fc}
(b) GDP
(c) NDP_{fc}
(d) Personal Income

7. Market value of all the final goods and services including depreciation is known as:

(a) GDP_{mp}
(b) GDP_{fc}
(c) NDP_{fc}
(d) Personal Income

8. GNP_{mp} =?

(a) GDP_{mp} - Depreciation
(b) GDP_{mp} + Net factor income from abroad
(c) (c) GDP + Depreciation
(d) none of these

9. Which of the following is not included in national disposable income?

(a) Net indirect tax
(b) Net factor income from abroad
(c) Net foreign transfer income
(d) Net Govt. transfer income

10. NNPmp =?

(a) GNP_{mp} - Depreciation
(b) GNP_{mp} + Depreciation
(c) GNP_{mp}+ Net indirect tax
(d) None of these

11. Which of the following is true?

(a) GNP = GDP + Depreciation
(b) NNp = GNP + Depreciation
(c) NNP = GNP - Depreciation
(d) GNP = NNP - Depreciation

12. Which of the following is not included in private income?

(a) Transfer earning
(b) Net factor income from abroad

(c) Government income (d) Interest from public debts

13. Net transfer income from rest of world is not included in:

(a) national income (b) private income
(c) personal income (d) personal disposal income

14. Which of the following does not include corporate taxes?

(a) Private income (b) National income (c) Personal income (d) GDP

15. Which of the following does not include direct taxes?

(a) National income (b) Private income
(c) Personal income (d) Personal disposable income

16. Which of the following does not include net indirect taxes?

(a) NNP_{mp} (b) National income
(c) NDP_{mp} (d) GNP_{mp}

17. Which of the following does not include net indirect taxes?

(a) Personal income (b)Private income
(c) National income (d) Personal disposable income

18. Consumption + Saving =?

(a) Personal income (b) National income
(c) Personal disposable income (d) GDP

19. Which of the following is not included in operating surplus?

(a) Rent and Royalty (b) Interest (c) Profit (d) Bonus

20. Expenditure on road construction is which type of expenditure?

(a) Income from sale of share (b) Uniform from employer
(c) Transfer payments (d) Advocate fees

21. Which of the following is included in depreciation?

(a) Gross domestic product (b) National income
(c) Net domestic product (d) Private income

22. Which of the following is not included in NDP_{fc}?

(a) Compensation of employee (b) Net factor income from abroad
(c) Operating surplus (d) Mixed income

23. Which of the following is considered as an indicator of economic growth?

(a) Increase in real per capita income (b) Increase in GDP
(c) Increase in GNP (d) Any of these

24. Which of the following is not included in personal disposable income?

(a) Factor income (b) Rent
(c) Operating surplus (d) Transfer income

25. Which of the following is not included in national income?

(a) Amount received from lottery won ticket (b) Windfall gains
(c) Amount non-won in gambling (d) All of these

26. Which of the following is a part of national income?

(a) Beggar's income (b) Income from tuition
(c) Dividend on share (d) None of these

27. Which of the following statements is true?

(a) Private income - Personal income + Corporate tax + Undistributed profit
(b) Personal disposable income - Personal income - Direct taxes
(c) GDP_{fc} GDP_{mp}-Net indirect taxes (d) All of the above

28. Which of the following services is not included in national income?

(a) Services of watchman
(b) Services of wife
(c) Services of advocate
(d) Services of doctor

ANSWERS

1. (a)	2. (d)	3. (a)	4. (c)	5. (c)	6. (a)	7. (a)	8. (b)	9. (d)	10.(a)
11.(c)	12.(c)	13.(a)	14.(c)	15.(d)	16.(b)	17.(c)	18.(c)	19.(d)	20.(a)
21.(a)	22.(b)	23.(a)	24.(b)	25.(d)	26.(c)	27(d)	28.(b)		

D. ASSERTION-REASON TYPE

Read the following statements-Assertion (A) and Reason (R), and select the correct alternative in each case:

(a) (A) is true, but (R) is false.
(b) Both (A) and (R) are true.
(c) Both (A) and (R) are true and (R) is the correct explanation of (A).
(d) Both (A) and (R) are true but (R) is not the correct explanation of (A).

1. Assertion (A) : Leakages like savings and taxes reduce the flow of income.
Reason (R) : Leakages are withdrawals from the circular flow.

2. Assertion (A) : Transfer payments are not included in the national income.
Reason (R) : Transfer payment is the income received by the people without rendering any productive work.

3. Assertion (A) : Net Factor Income from abroad is the difference between factor income earned from abroad and factor income paid to abroad.
Reason (R) : When NFIA added to the domestic income it becomes national income.

4. Assertion (A) : Double counting leads to the inflated national income.
Reason (R) : When value of the same product added in the national product more than once is known as double counting.

5. Assertion (A) : Operating surplus includes rent, interest and profit.
Reason (R) : Operating surplus is the income from property and entrepreneurship.

ANSWERS

1. (a)	2. (a)	3. (b)	4. (b)	5. (b)

2 MARK QUESTIONS

1. What is national income?

Ans. The aggregate value of all final goods and services produced by the residents of a country, operating both within the national boundary and abroad, during any particular year, is called the national income of the country.

2. What do you mean by domestic territory?
Ans. It refers to the geographical or political boundary of the country excluding foreign embassy and international institutions like U.N., W.H.O., offices located within the geographical territory and including the embassy of this country located outside its geographical territory.

3. Who is known as normal resident?
Ans. Normal residents are those people who ordinarily reside in a country in which they live and whose economic interest lies in that country. They may or may not be the citizen of the country. It means normal residents include nationals as well as foreign nationals.

4. Define GDP.
Ans. The sum of the values of all final goods and services produced within the national boundary of a country during a particular year is called the Gross Domestic Product (GDP) (at market prices).

5. What do you mean by GNP?
Ans. The sum of the values of different goods and services produced by domestically owned factors of a country during any particular year, is called the Gross National Product (GNP) (at market prices).

6. What is domestic income?
Ans. Domestic income of a country refers to the income earned by the residents and non-residents within the domestic territory of the country during any particular year.

7. What do you mean by GDP at factor cost?
Ans. It is the sum total of factor income earned by all factors of production for producing goods and services within a financial year.

8. Show the relationship between GDP_{fc} and GDP_{mp}
Ans. GDP_{fc}=GDP_{mp}-(Indirect taxes - Subsidy)
Where, GDP Gross domestic product (at factor cost), and
GDP_{mp}=Gross domestic product (at market price).

9. Show the relationship between NNPfe and NNPmp
Ans. NNP_{fc} = NNP_{mp}-(Indirect taxes - Subsidy)
NNP_{fc} = NNP_{mp}-NIT

10. What do you mean by private income?
Ans. The private income refers to the factor incomes earned within the domestic territory and abroad by the factor owners in the private sector (of a country) including the current transfers from the Government and net transfers from the rest of the world.

11. What is personal income?
Ans. The personal income refers to the current income of persons or household (of a country) from all sources during any particular year. Personal income = private income- undistributed profit - corporate profit tax - retained earnings of foreign companies.

12. What do you mean by national disposable income?
Ans. The national disposable income refers to the income available to different households, business firms and the Government of a country for spending during any particular year. It is defined as the sum total of national income at market price and current transfers received from rest of the world.

13. What is nominal GDP?
Ans. When the GDP of a country is valued at current market price during any year, it is called nominal GDP.

14. What do you mean by real GDP?
Ans. When the GDP of a country during any year is measured at some constant or base year price, it is called real GDP.

15. Define per capita income.
Ans. Per capita income denotes the average income of different individuals in an economy during any particular year.

Per capita income = National income / Total population of the country

16. What is per capita real income?
Ans. Per capita real income denotes the average income of different individuals of an economy during any particular year, in real terms. It actually implies the purchasing power of the money income of an individual. When national income is measured in terms of constant prices, we get real national income.

$$Per\ capita\ real\ income = \frac{real\ national\ income}{Total\ population\ of\ the\ country}$$

17. What is the solution to the problem of double counting in the estimation of national income?

Ans. The problem of double counting in the national income estimation can be solved in two ways:
(i) Only the value of final goods has to be taken into account, or
(ii)The value-added at each stage of production has to be estimated, where
Value-added = Value of output - Value of raw materials.

18. What is double counting and what does it lead to?
Ans. The error which arises in estimating the nation's output by just adding the values of output of all firms, is known as the error of double counting. For instance, if we estimate the values of both bread and flour, this error will arise because the value of flour has already been included in the value of bread. So, the value of flour will be counted twice. This error of double counting would lead to an exaggerated figure of the nation's output. (Inflated National Income)

19. Why are pure exchange transactions excluded from national income accounting?
Ans. All types of 'pure exchange transactions' are excluded from national income accounting. In case of pure exchange transactions nothing new is produced in the current year. For instance, secondhand sales, purchase and sale of securities (shares and debentures), transfer payments (such as unemployment allowances, pension payments), etc., are regarded as pure exchange transactions. All such transactions are not concerned with current year production. So, they are excluded from nations income estimates.

20. What is operating surplus?
Ans. The operating surplus is defined as the excess of value added of the producers over the sum of intermediate consumption, mixed income of self-employed, compensation of employees, net indirect taxes and depreciation allowance for fixed capital.
Operating surplus = Gross value added at market prices-(Intermediate consumption + Compensation of employees + Mixed income self-employed + Net indirect taxes + Depreciation allowance).

Or

It is the income earned from property and entrepreneurship.
Operating surplus = Property income + Income from entrepreneurship
=Rent + Interest + Profits.

21. What do you mean by compensation of employees?
Ans. The payments of wages and salaries made by the producers to their employees, and the contribution of those producers to the social security schemes in respect of their employees (e.g., contribution towards insurance premium), are termed ascompensation of employees in national income accounting.

22. Define transfer payments.
Ans. The payments received by the households, business firms and non-profit organisation from the Government and other sources without rendering any productive activities, can be considered as transfer earnings. The old age pension, unemployment benefits, scholarship of poor students, etc., are the examples of such transfer earnings.

23. Give examples of transfer income of the Government of a country.
Ans. The Government receives transfer income from the households and enterprises in the form of direct taxes (say, income tax) and indirect taxes (say, excise duty). Since these receipts do not add anything to the current flow of goods and services in the economy, they are considered as transfer income.

24. What is domestic private final consumption expenditure?
Ans. The final consumption expenditure made by the resident households and private non-profit institutions (who serve households) within the domestic territory of a country during any particular year, can be considered as the domestic private final consumption expenditure.

25. By which official agency is the estimation of national income in India done?\
Ans. The estimation of national income is done by the Central Statistical Organisation (CSO).

26. What do you mean by the circular flow of income?
Ans. Circular flow of income is defined as the flow of payments and receipts for goods and services and factor services among the different sectors of the economy.

27. What is depreciation?
Ans. The loss of value of capital goods due to its wear and tear in the process of production either due to normal wear and tear or due to obsolescence.

28. What two precautions need to be taken while calculating national income through the expenditure method?
Ans. Two precautions in the expenditure method:
(i) Expenditure on financial assets like shares and bonds is excluded because it reflects only the transfer in the ownership of these assets.
(ii) All expenditures on secondhand goods should be excluded because national income measures the value of only the currently produced goods and services.

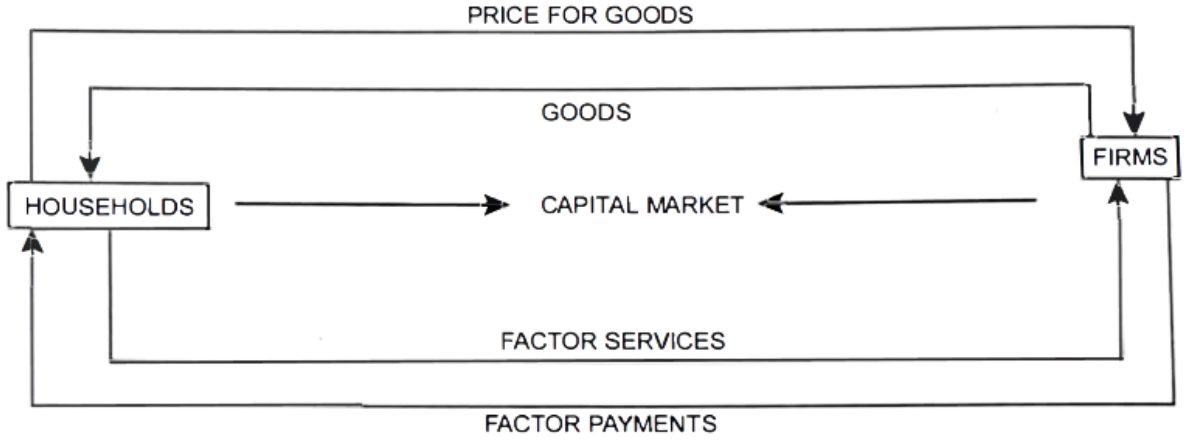
29. What do you mean by leakages in the circular flow?
Ans. Leakages means outflow of such an amount of fund from the circular flow that leads to a contraction in the national income of the country. For examples: savings, taxes, imports, etc.

30. What do you mean by injections into the circular flow?
Ans. It refers to such an amount of inflow of fund into the circular flow that causes an expansion in national income of a country. For examples: investment, exports, subsidies, etc.

31. Give any two conceptual difficulties faced in the measurement of national income.
Ans. **(i) Difficulty of Defining Nation**: According to the definition of 'nation' as used in the studies of national income accounting, national income doesn't only include income produced within the country but also income earned in other countries by way of shipping charges, interest payments, insurance and banking, minus any payment made to foreign countries.
Therefore, the definition of nation goes beyond the political boundaries.
(ii) Choice of Method: Another problem relates to the choice between different methods in the calculation of national income. However, there is an agreement among all economists to use these methods simultaneously depending upon the availability of data.

3-4 MARKS QUESTIONS

1. Explain circular flow of income in two - sector economy.
Ans. Circular flow implies the flow of goods and services or the flow of money across the different sectors of economy. Each flow shows how one sector depends on the other:



From the above model we get the following observations:
(i) Total production of goods and services by firm equals to total consumption of goods and services by household sector.
(ii) Factor payments by firm equals to factors incomes by household.
(iii) Consumption expenditure of household equals to income of the firm.
(iv) Real flow of production and consumption of firms and household equals to money flow of income and expenditure of firm and household.
(v) From the above observation it is clear that the circular flow of money will remain constant and unchanged which is known as equilibrium in circular flow of income.
In real world entire income is never spent on consumption, some part of it is saved. All savings done by household are deposited in financial system, their savings return to the firms for investment. Thus, this two-sector model gives the following additional informations:

(i) All income is not converted into consumption expenditure, some part of it is saved. Thus,

$$Y=C+S$$
(Income = Consumption + Savings)

(ii) S is converted into I (investment) which is the expenditure done by the producer. Thus,

(Savings = Investment)

(iii) Y=C+S and S = 1. Thus, we can conclude by saying that

$$Y=C+1$$

In this model, the condition for equilibrium is S = I or C + S = C + I.

If S>I then the income flow decreases because 5 is known as withdrawal or leakage in circular flow. If S <I then the income flow increases because Investment is known as injection in circular flow.

2. What is meant by net factor income from abroad? Explain its components.
Ans. Net factor income from abroad may be defined as the income received from abroad for rendering factor services abroad and income paid for the factor services rendered by non-residents in the domestic territory of a country.
The components of net factor income from abroad are:
(i) Net Compensation of the Employees: It is equal to the difference between the compensation of employees received by resident workers that are living and employed abroad and similar payments made to non-resident workers.
(ii) Net Income from Property and Entrepreneurship: It is equal to the difference between the net income received by way of interest, rent and profits by the residents and similar payments made to the rest of the world.
(iii) Net Retained Earning from Abroad: It is equal to the difference between retained earning of the foreign countries in the country and retained companies located abroad.

National income - Domestic factor + Net factor income from abroad

3. What precautions should be taken to measure the national income by output method?

Ans. **Precautions:**

S.No.	Things to be added	Things to be excluded
1.	Value of goods for self consumption.	Sale and purchase of secondhand goods.
2.	Imputed value of owner occupied houses.	Value of intermediate goods.
3.	Own a/c production of fix capital by households, firms and Government.	Sale and purchase of bonds and shares because they only reflect the transfer of funds, not an addition to N .I. Domestic services like services of housewives.

4. Give the steps to measure national income by value added method.

Ans. **Steps:**

(i) Identifying production units and classifying them into various industrial sectors (primary, secondary, tertiary).

(ii) Estimating net value added at FC (only final goods to avoid double counting):

(a) Estimating gross.

(b) Deduct the intermediate product and depreciation.

(c) Deduct NIT from MP.

(iii) Estimating net domestic output (Sum total of net value added at FC of all the industrial enterprises in the domestic territory of a country).

(iv) Estimating net factor income from abroad.

5. Complete the following equations:

(i)	GNP	=	NNP+
(ii)	NNP	=	NDP+
(iii)	GDP_{mp}	=	GDP_{fc}+
(iv)	GDP	=	NDP+
(v)	NI at factor cost	=	NI at market price-
(vi)	DPI	=	PI-

Ans.

(i)	GNP	=	NNP + Depreciation.
(ii)	NNP	=	NDP+ Net factor income from abroad.
(iii)	GDP_{mp}	=	GDPfc + Net indirect taxes (NIT).
(iv)	GDP	=	GDP = NDP + Depreciation
(v)	NI at factor cost	=	Ni at market price - NIT.
(vi)	DPI	=	PI-Income tax.

6. Why the GDP measure may be higher than the GNP measure in any country?

Ans. The GDP measure in any country may be higher than the GNP measure because of negative net property income of the factors earned abroad. We know that:

GNP = GDP + net property income of factors earned abroad (say, NFIA).

If NFIA <0 then GNP <GDP.

It implies that the factor incomes earned abroad by the citizens of that country are less than the factor incomes paid abroad.

7. Mention four constituents of the final consumption expenditure made by the resident households in a country.

Ans. The following are the four constituents of the final consumption expenditure made by the resident households in a country:

(i) Expenditure on new durable and non-durable goods and services.

(ii) Direct purchases of consumer goods made abroad, imports.

(iii) Wages, salaries and other benefits received in kind.

(iv) Food and other items produced for self-consumption.

8. Explain the importance of circular of income.
Ans. Circular flow models are important because they:
(i) show interdependence between sectors,
(ii) provide knowledge of structure of the economy,
(iii) facilitate the estimation of national income,
(iv) give information on important macro variables, and
(v) give information about injections and withdrawals

9. Give the importance of real national income.
Ans. Advantages of real national income:
(i) It reflects the real change in the volume of production of goods and services, hence it is a true indicator of economic progress of the country.
(ii) It enables us to make year-wise comparison of changes in the growth of output of goods and services.
(iii) It is often used in making international comparisons.

10. Mention three methods of measuring national income.
Ans. We can mention the following three methods of measuring national income:
(i) the product method (also known as the value-added method), (ii) the income method and (iii) the expenditure method.

(i) In the product method we first estimate Gross National Product (GNP) by summing up the values added by all the producers in the economy. We then subtract depreciation of capital stock and indirect business taxes from GNP to arrive at national income.

(ii) In the income method, we add up the incomes of all the income-earning units in the economy.
(iii) Since every income comes from an act of expenditure, national income can also be obtained by calculating national expenditure, i.e., the sum of all expenditure items in the economy.

11. Indicate two factors which make the measurement of national income difficult in any less developed country like India.
Ans. Among the many factors that make the estimation of national income difficult in a country like India, we may mention the following two:
(i) The Indian economy is predominantly agricultural. Estimation of agricultural income is a difficult proposition due to several reasons. For instance, a part of agricultural production is consumed by the farmers themselves. This part is not recorded as a part of national income (while, in fact, it should be recorded as such).
(ii) People in an underdeveloped country like India are not accustomed to keeping proper record and accounts of their economic activities. This makes the task of collecting data regarding the value added by the producers quite difficult.

12. Name three components of the domestic factor income.
Ans. Three components of the domestic factor income are:
(i) Compensation of employees (e.g., wages and salaries, employer's contribution to social security schemes, etc.). This is known as labour income also.
(ii) Operating surplus (viz., rent, interest and profits). It is the income earned from property and entrepreneurship.
(iii) Mixed income of self-employed persons, e.g., income of a doctor running a clinic.

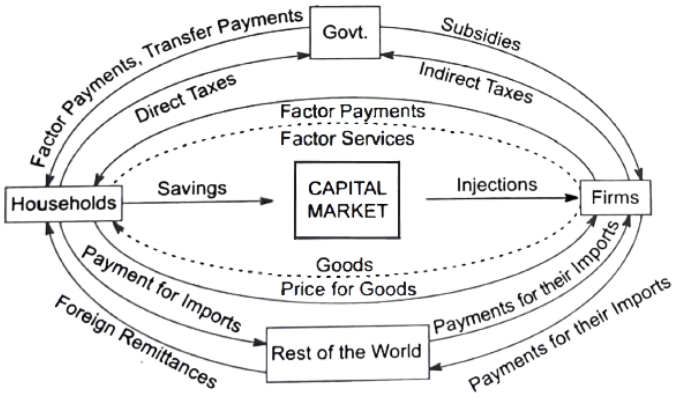
13. Will the following be included in gross domestic product/domestic income of India? Give reasons for each answer.
(i) Consultation fee received by a doctor.
(ii) Purchase of new shares of a domestic firm.
(iii) Profits earned by a foreign bank from its branches in India.
(iv) Service charges paid to a dealer (broker) in exchange of secondhand goods.
Ans. (i) Yes, it is a factor income. It is his salary.
(ii) No, it is not included in GDP, because it is a merely financial transaction which does not help directly in production.
(iii) Yes, it is a factor income in domestic territory.
(iv) Yes, it is included because it is his factor income (salary).

14. Give the difficulties faced during measurement of national income by output method.
Ans. **(i) Valuation of inventories**: It is difficult to calculate inventories because to decide whether to calculate it at current prices or base year prices because the value on basis of base year prices will be different from the value of current year prices.
(ii) Estimation of depreciation: It is difficult to decide whether to estimate depreciation at current year prices or base year prices. There are different methods to calculate depreciation. It is difficult to choose an appropriate method for the measurement of depreciation.
(iii) Concept of N.I.: Some concepts of N.I. are not clear at all because it is difficult to say whether education, transport, medical services should be included in final goods or intermediate goods. On the other hand, few countries do not include services in N.I. this also creates lot of confusion in the measurement of N.I.
(iv) Difference between intermediate and final goods: It is very difficult to draw a clear cut difference between intermediate and final goods because same goods can be treated as intermediate and final goods on the basis of their purpose.

15. State whether the following is a stock or flow:
(1) National capital, (ii) Exports, (iii) Capital formation, and (iv) Expenditure on food by households.
Ans. Stock: (i), since national capital is a variable measurable at a point of time.
Flow: (ii), (iii) and (iv), since these are variables measurable over period of time.

6-8 MARKS QUESTIONS

1. Explain circular flow of income in a four-sector economy. (Open economy)
Ans. Circular flow implies the flow of goods and services or the flow of money across the different sectors of economy. Each flow shows how one sector depends on the other. There are four sectors in an open economy.



Four-sector model is known as the model of open economy. Open economy means an economy which has international relations. In this model all the four sectors of model; household, firms, Government and rest of the world will be operating. We get details given ahead from the above observations:

(i) Total production of goods and services by firm equals to total consumption of goods and services by household sector.

(ii) Factor payments by firm equal to factor incomes by household.

(iii) Consumption expenditure of household equals to income of the firm.

(iv) Real flow of production and consumption of firms and household equals to money flow of income and expenditure o firm and household.

(v) From the above observation it is clear that the circular flow of money will remain constant and unchanged which is known as equilibrium in circular flow of income.

In real world entire income is never spend on consumption, some part of it is saved. All savings done by household are deposited in financial system and their savings return to the firms for investment.

(vi) Government sector receives direct taxes from the household sector and the indirect firm.

(vii) Government sector makes factor payments and transfer payments to the households and give subsidies to the business sector.

(viii) Rest of the world sector makes payment to the firm for their imports.

(ix) External sector receives payments from the firms from their exports.

(x) Households export its man-power to external sector.

(xi) Households get foreign remittances from rest of the world.

(xii) Households make payment to the foreign sector for whatever import.

In the above flow leakages are savings, taxes and imports. Injections are investment, Government expenditure and exports.

Thus,

Savings + Taxes + M (Imports) = Investment + Government expenditure +X (Exports)

Then economy will be in equilibrium.

2. What precautions must be taken while estimating national income by income method?

Ans. The following precautions should be taken into consideration while estimating national income by income method:

(i) Transfer payment should not be included.

(ii) Value of the production for self-consumption and imputed rent of owner occupied.

(iii) Houses have to be included.

(iv) Illegal incomes should not be included.

(v) Windfall gains should not be included.

(vi) Death duties, gifts tax, wealth tax and tax on windfall gains are paid from the past savings or wealth, therefore these are not a part of national income.

(vii) While calculating national income profit before deduction of corporation tax will not be included.

(viii) If a person receives money by selling some secondhand goods that amount will not be included.

3. Give the steps to measure national income by income method.

Ans. **Steps:**

1. Classification of sectors

2. Classification of various factor incomes like:

(i) Compensation of Employees: Wages, salaries and other payments like pensions.

(**ii) Operating Surplus**: Income paid for the ownership and control of the capital. It includes rent, interest, royalties, dividends, undistributed profits of corporations before corporation taxes and profits of the Government enterprises.

3. Mixed Income: Labour income and capital income of those who provide both labour and capital services in the production process.
(i) Domestic factor income is estimated.
Domestic factor income or NDP Compensation of employees + Operating surplus + Mixed income
(ii) Estimation of net factor income from abroad
National Income or NNP = NDP + NFIA

4. Give the steps to measure national income by expenditure method.
Ans. **Steps:**
1. Division of all those Economic Units which Incur Expenditure on Final Product:
(i) Household, (ii) Business sector, (iii) Government sector, (iv) Rest of the world.
2. Division of Final Expenditure:
(i) Consumption expenditure, (ii) Investment expenditure, (iii) Government expenditure, (iv) Net export.
3. Measurement of Components of Final Expenditure:
(i) Private Final Consumption Expenditure: Expenditure on purchase of goods and services divided into three Categories:
(a) Expenditure on non-durable goods such as food.
(b) Expenditure on durable goods like car.
(c) Expenditure on services like transport services.
To estimate private final consumption expenditure two types of data are required: (1) volume of sales in the market, (II) retail prices.
Precautions while Calculating Private Final Consumption Expenditure:

(i) Purchase of new house is not included.

(ii) Second-hand goods and their sale and purchase are not included.

(iii) Production for self-consumption will be included.

(iv) Imputed rent will also be included.

(ii) Estimation of Investment Expenditure: There are three categories of investment expenditure:
(a) Expenditure on business fixed investment.
(b) Inventory investment.
(c) Residential investment.
Depreciation will also be calculated and deducted from gross investment expenditure.
Net Domestic Investment = Gross fixed business investment + Inventory investment + Residential investment - Depreciation
(iii) Government Final Consumption Expenditure: Government final consumption expenditure is the sum total of the following items:
(a) Compensation of employees paid by the Government.
(b) Goods and services purchased by the Government from domestic market(intermediate goods).

(c) Purchases from abroad.
Government sales are excluded from it and transfer payments are not included in it because they do not

contribute anything into the current production.

(iv) Estimation of Net Exports: Net exports is the difference between exports and imports of a country during the period of one year. Net export is a part of domestic product and this may be both positive as well as negative. It is also called net foreign investment because it increases wealth and productive capacity of the country. Therefore, the value of export should be added to expenditure of the normal residents. Likewise the value of imports are a part of the domestic product of the foreign country and thereby they have to be deducted from the value of exports to arrive at net exports.

4. The sum total of all the four items give NDP_{mp} By deducting NIT from it, we get NDP_{fc} .

5. Estimation of net factor income from abroad.

$$\text{National income} = C + In + G + (X\text{-}M) - NIT + NFIA$$

5. Explain the limitations of real per capita income as a measure of economic welfare.

Ans. GDP is used as an indicator of economic welfare. However, it is not a satisfactory measure of economic welfare due to following reasons:

(i) **Composition of GDP**: GDP shows the total of goods and services produced in the country. However, it does not exhibit the structure of the total product. An increase in GDP can be mainly due to increased production of war goods, instead of consumer goods. Such an increase will not be associated with any improvement in economic welfare.

(ii) **Changes in Prices**: If increase in GDP is due to rise in prices and not due to increase in physical output, then it will not be a reliable index of economic welfare.

(iii) **Non-Monetary Exchanges**: Correct estimates of GDP are not available in the most of the under-developed countries. Several exchange transactions are still conducted in the rural areas without the use of money. Such transactions are never recorded in the estimation of GDP. Under-estimated GDP cannot be considered as the true index of economic welfare.

(iv) **Uneven Distribution of GDP**: Increase in GDP will not promote welfare if it is unevenly distributed among the people. Disparities in income distribution only benefit the rich and poor are denied the benefits of the increase in GDP.

QUESTIONS BASED ON DIFFERENTIATION

1. Distinguish between factor income and transfer income.

Ans.

S.No.	Basis	Factor Income	Transfer Income
1.	**Meaning**	It refers to the income accuring to the owners of factor of production.	It refers to the income received without rendering any service.
2.	**Nature**	It is earned income as it is received in lieu of factor services.	It is an unearned income as it is received without rendering any productive services.
3.	**Examples**	Rent, wage, interest and profits.	Scholarship to students, unemployment allowances, old age pension, pocket money, etc.

2. Distinguish between personal disposable income and national disposable income.

Ans.

S.No.	Basis	Personal Disposable Income	National Disposable Income.
1.	**Meaning**	It is sum of factor and non- factor incomes of households after payment of direct taxes.	It is the sum of NNP_{mp} and net current transfers from the rest of the world.
2.	**Nature**	PDI = Households consumption + Households savings.	NDI = National consumption expenditure + National saving

3. Distinguish between net exports and net factor income from abroad.

Ans.

S.No.	Basis	Net Exports	Net Factor Income.
1.	**Meaning**	It refers to difference between exports and imports of goods and non-factor services.	It refers to difference between factor income from abroad and factor income to abroad.
2.	**Nature**	It is domestic concept.	It is a national concept.
3.	**Examples**	It includes non-factor services.	It includes factor services.

4. Distinguish between national disposable income and national income.

Ans.

S.No.	Basis	Personal Disposable Income	National Disposable Income.
1.	**Nature of Income**	It includes factor income as well as transfer income.	It refers to the factor incomes received by the residents of a country.
2.	**Estimation**	It is estimated at market price i.e., it includes net indirect taxes	It is estimated at factor cost, i.e., it excludes net indirect taxes

5. Differentiate between national income at constant prices and national income at current prices.

Ans.

S.No.	Basis	Net Exports	Net Factor Income.
1.	**Meaning**	When N.I. is measured at some base year price is called N .I. at constant price a real national income.	When N.I. is measured at prevailing market price is known as N.I. at current prices.
2.	**Causes of change**	It only gets affected by changes in physical output	It is affected by: (i) change in price, (ii) change in physical output of goods and services.
3.	**Examples**	It is a true indicator of economic growth	It is not a true indicator of economic growth of a country.

6. Distinguish between GDP and GNP

Or What is the difference between domestic product and national product?

Ans. The total output produced in the economy, measured by the GDP, differs from total income received in an open economy, which is measured by the GNP.

Thus, GNP (at market prices) = GDP (at market prices) +NFIA (Net factor income from abroad)

Hence, a part of GNP is earned abroad, but GDP does not include such activities because, it represents only those activities which lead to production of final goods and services within the domestic territory of the country.

7. How is national income different from GNP at market prices?

Ans. National income of a country may be defined as the Net National Product (NNP) at factor cost, and this is calculated as follows:

GNP (at market prices) - Depreciation cost of capital NNP (at market prices)-NIT=NNP (at factor costs) = National income.

8. Differentiate between personal income and national income at current prices.

Ans.

S.No.	Basis	Net Exports	Net Factor Income.
1.	**Meaning**	It is the sum total of all incomes that are actually received by households from all sources.	National Income refers to the sum total of all the factor incomes, earned by the normal residents of a country during a period of one year.
2.	**Nature of income**	It includes both factor incomes as well as transfer income.	It includes only the factor increases.
3.	**Public Sector Income**	It does not include income earned by public sector.	It includes income earned by public sector.
4.	**Formula**	Personal Income = Private income – Corporate tax – Retained earnings	$NNP_{fc} = GNP_{fc}$ - Depreciation

9. Distinguish between personal income and personal disposable income.

Ans. Disposable income means the income available to the citizens of the country for the purpose of consumption or saving.

This is obtained by deducting direct taxes paid from personal income:

Disposable income = Personal income - direct taxes

On the other hand,

Personal income = National income - Undistributed corporate income + Transfer payments.

10. What is the difference between GNP at factor cost and GNP at market prices?

Ans. The difference between GNP at factor cost and GNP at market prices is equal to net indirect taxes (NIT). The indirect business taxes are included in the market price but they do not flow to the factors as factor incomes. However, market price does not include subsidy payments by the Government but it flows to the factors as factor payments.

GNP (at factor cost) = GNP (at market price) - (Indirect taxes - Subsidy).

11. What is the distinction between NNP at factor cost and at market price?

Ans. The NNP valued at market price can be determined by subtracting the depreciation cost from the GNP valued at market price. However, NNP at factor cost actually implies the national income (ie., the sum total of factor incomes like wages, rent, interest and profit). The following adjustments are required to determine NNP at factor cost from NNP at market price:

NNP (at market price) - Indirect taxes + Subsidy payments = NNP (at factor cost).

12. What is the difference between GNP_{mp} and NDP_{fc}?

Ans. The gross national product (GNP) at market price measures the total values of final goods and services produced by domestically owned factors of a country during any year. This measure includes the net factor income earned abroad. On the other hand, the net domestic product (NDP) at factor cost measures the values of final goods and services produced within the national boundary of the country in terms of factor incomes. This measure excludes the depreciation cost of capital.

13. Indicate any two differences between the national income and domestic income.

Ans. (i) The national income refers to the incomes earned by the residents earned both within and outside the domestic territory of a country during any year. The domestic income, however, refers to the incomes earned by both residents and non-residents within the domestic territory of a country during any year.

(ii) The national income includes net factor income earned abroad, while the domestic income does not include this item.

14. Mention any two differences between the final goods and intermediate goods.

Ans. (i) The final goods are only used for final consumption (e.g., coal used at home for cooking purposes). However, the intermediate goods are used for producing other goods (e.g., coal used in a factory).

(ii) The final goods are included in the calculation of national income. But the value of intermediate goods is not included in national income estimation.

15. Distinguish between gross and net value added.
Ans. If we deduct the value of intermediate consumption from the value of output, we get gross value added. However, when we deduct the depreciation allowance for fixed capital from gross value added, we get net value added.

16. Distinguish between value added at market price and value added at factor cost.
Ans. The value added at factor cost refers to the addition made to the value of inputs by the contributions of various factors of production. However, if such value added is measured in terms of market prices, we call it value added at market prices. In fact, value added at market prices: Indirect taxes + Subsidy - Value added at factor cost.

17. Distinguish between personal income and private income.
Ans.

S.No.	Basis	Personal Income	Private Income
1.	**Meaning**	It refers to the income which accrues to households from all sources.	It refers to the income which accrues to private sector from all sources.
2.	**Concept**	It is a narrower concept as it is a part of private income.	It is a broader concept as it includes personal income.
3.	**Formula**	Personal Income = Private income - Corporate tax- Retained earnings.	Private income Personal income + Corporate tax + Retained earnings.

18. Distinguish between current transfer and capital transfer payments.
Ans. The transfer payments for the consumption purposes of the recipients and made out of the current income of the payer, can be considered as the current transfer payments. For example, the old age pension given by the Government to the retired employees.
On the other hand, the unilateral transfer payments made by the individuals, enterprises and the Government from their past savings for facilitating long-term investment (or gross capital formation) of the recipient, can be regarded as capital transfer payments. For instance, payments made by the Government to the flood-striken households for the reconstruction of their damaged house buildings.

19. Differentiate between stock and flow.
Ans.

S.No.	Stock	Flow
1.	It means the quantity of economic variable which is measured It means the quantity of economic variable which is measured at a particular point of time.	It means the quantity of economic variable which is measured during a period of time.
2.	Have no time dimensions.	Has time dimensions as per hour, per day, per month and per year.
3.	Static concept.	Dynamic concept.
4.	For examples: Wealth, property, quantity of money, etc.	For examples: Income, consumptions, investment, water in a river, etc.

CASE BASED QUESTIONS

CASE 1

National income is the monetary value of all final goods and services produced by the normal residents of a country in a year. The national income of a country is measured to know the economic growth of the country and it is used to compare the economic condition of one country with the other. There are many aggregates of national income which are used to do the comparative study of different economies.

Answer the following questions on the basis of the above paragraph:

1. What are final goods?
2. Why only final goods included in the national income?
3. Differentiate between final goods and intermediate goods.
4. Is national income a stock or flow?
5. Who is known as a normal resident?

Ans. 1. Final goods are those goods that are used by the consumers and have no use in the future.

2. To avoid the problem of double counting.

3. (a) The final goods are only used for final consumption (e.g., coal used at home for cooking purposes). However, the intermediate goods are used for producing other goods (e.g., coal used in a factory).

(b) The final goods are included in the calculation of national income. But the value of intermediate goods is not included in national income estimation.

4. National income is a flow as it is measured over a period of time i.e., one year.

5. Normal residents are those people who ordinarily reside in a country in which they live & whose economic interest lies in that country. They may or may not be the citizen of the country. It means normal residents include nationals as well as foreign nationals.

CASE 2

Gross Domestic Product measures the aggregate production of final goods and services taking place within the domestic country in a year. But there are many citizens of India who are working abroad, and their income is not included in it. At the same time there are many foreign citizens who work in India and their income earned over here will not be added in the national income of India. In order to get the National Income of the country we have to add and subtract many macro-economic variables in it.

Answer the following questions on the basis of the above paragraph:

1. Which aggregate of national income is technically known National Income?
2. What is NFIA?
3. How to get National Income from GDP_{mp}?
4. What is Depreciation?

Ans. 1. NNP [net national product at factor cost]

2. Net factor income from abroad may be defined as the difference between the income received from abroad for rendering factor services abroad and income paid for the factor services rendered by non-residents in the domestic territory of a country.

3. GDP_{mp}-Depreciation + NFIA- NIT = NNP_{fc}

4. Depreciation is a reduction in the value of fixed assets due to normal wear and tear.

CASE 3

Flow of goods and money across different sectors of the economy is known as circular flow. As we all know that one man's expenditure is other man's income. Households spend money on buying goods which are manufactured by the producers and that money becomes producer's income which they spend on hiring factors of production and giving taxes to the government. The money which government gets in the form of taxes uses to give subsidies to the firms and factor payments and transfer payments to the households. This process of give and take keeps on moving like this forever.

Answer the following questions on the basis of the above paragraph:

1. Explain two types of flows in the economy.
2. Which economy is known as closed economy?
3. According to the above paragraph what will be the leakage and injection in the flow and why?
4. Explain the role of leakages and injections in the circular flow.
5. Draw a well labelled diagram of circular flow of two sector model.

Ans. 1. There are two types of flows of a circular flow. Real and Money.

Real flow is a flow of goods and services across different sectors of the economy.

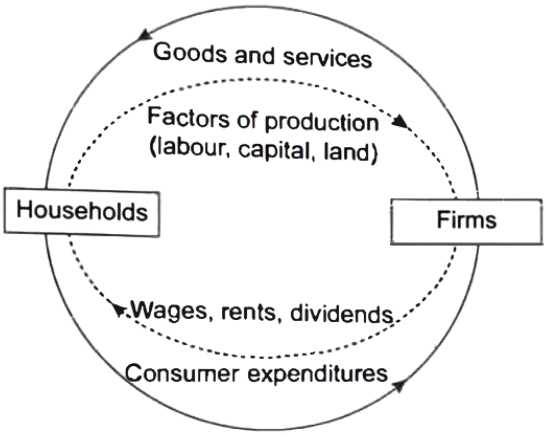
Money flow is the flow of income or money across different sectors of the economy.

2. Economy which does not have international relations is known as closed economy.

3. Tax is a leakage as it reduces the flow of money. Subsidy is an injection as it increases the flow.

4. Leakages reduce the flow of money and injections increase the flow.

5.



QUESTIONS WITH HIGH DIFFICULTY

1. While estimating national income how will you treat the following?
(i) Imputed rents of self-occupied houses;
(ii) Financial help received by flood victims; and
(iii) Interest received on debentures.
Ans. (i) The imputed rents of self-occupied houses are included in national income accounting as the value of services generated during any year.
(ii) This item is not included in national income accounting since it is considered as transfer income.
(iii) This interest income will be included in national income accounting when the fund raised through the sale of debentures is used by the firm for production in the current year.

2. How will you treat the following while estimating national income?
(i) Capital gain on the sale of a house;
(ii) Prize won in a lottery; and
(iii) Interest on public debt.
Ans. (i) This capital gain should not be included in national income (N.I.) accounting since it does not correspond to the creation of new assets in the current year.
(ii) This item is not also included in N.I. accounting because it does not correspond to the production of any goods and services in the current year.
(iii) This interest income will not be included in the N.I. accounting when this fund is used by the Government for consumption purposes. (It is considered as transfers income of the recipient).

3. How will you treat the following while estimating national income of India?
(i) Salaries paid to the Russians working in the Indian embassy in Russia;
(ii) Profit earned by an Indian company from its branch located in Singapore;
(iii) Capital gains to Indian residents from the sale of shares of foreign companies.
Ans. (i) It is included in the N.I. accounting of India as factor payment paid abroad.
(ii) This item is included in the N.I. accounting of India as factor income earned abroad.

(iii) This item is not included in the N .I. accounting since it does not correspond to any value creation (or creation of goods and services) in the current year.

4. How will you treat the following while estimating national income (N.I.) of India?
(i) Salaries received by Indian residents working in Russian embassy in India.
(ii) Profits earned by an Indian bank from its overseas branches.
(iii) Entertainment tax received by the Government.
Ans. (i) It is included in the N.I. accounting of India as factor income earned abroad.
(ii) This item is also including in the N.I. accounting of India as factor income earned abroad.
(iii) The entertainment tax received by the Government is considered to be a transfer payments received by the Government from the households and it does not correspond to current flow of goods and services. So, it is not included in the N. estimation.

5. Are the following a part of country's NDP_{mp}?
(i) Net indirect taxes;
(ii) Net exports;
(iii) Net factor income from abroad; and
(iv) Consumption of fixed capital.
Ans. (i) The 'net indirect taxes' (NIT) is a part of the net domestic product (at market price) (NDP_{mp}) [but NIT is deducted from NDP_{mp} to calculate NDP at factor cost.]
(ii) The 'net exports' is a part of the NDP_{mp} when we calculate it on the basis of total expenditure method.
(iii) The 'net factor income from abroad' is not a part of NDP_{mp} [If we add this to NDP_{mp} we get NNP_{mp}].

6. How would you treat the following issues while estimating the national income?
(i) Payment of fees to a lawyer engaged by a firm;
(ii) Rent-free house to an employee by an employer; and
(iii) Purchases by foreign tourists.
Ans. (i) Since it is considered as an intermediate expenditure of the firm, it is subtracted from the sales value while calculating the gross value added of the firm [We know that GDP = GVA of all firms].
(ii) It is included in the national income estimation since it is a part of compensation of employees.
(iii) This is also included in national income estimation since it resembles exports of goods and services to the foreign purchasers.

7. State whether the following can be included in national income (N.I.) accounting:
(i) Transport expenses by a firm;
(ii) Expenditure on construction of a house;
(iii) Gift received from the employer;
(iv) Purchase of machine by a factory;
(v) Salary received by an Indian resident working in US embassy in New Delhi;
(vi) Interest paid on loan taken to purchase a personal car.
Ans. (i) It is an intermediate expenditure incurred by the firm and, therefore, it is deducted from the value of sales for estimating the gross value added of the firm [we know that GDP GVA of all firms].

(ii) This expenditure is included in N.I. accounting since it corresponds to the creation of goods during the current year.
(iii) This item is not included in N.I. accounting since it is considered to be a transfer income received by an employee from his/her employer.
(iv) The purchase of new machines by a factory would mean fixed capital formation during the current year and hence, it is included in N.I. accounting.
(v) It is included in N.I. accounting as 'factor income earned abroad".
(vi) This interest payment on loan is not included in N.I. accounting since it does not correspond to the creation of any goods in the current year.

8. State whether the following factor income would be treated as domestic income of India or not:
(i) Profits earned by foreign banks from their branches in India.
(ii) Salary received by Indian residents working in the American embassy in India.
(iii) Profits earned by an Indian company from its branch in Singapore.
(iv) Compensation of employees given to the residents of China working in Indian embassy in China.
Ans. (i) It is a part of the domestic income of India because the foreign banks (non residents) are located within the domestic territory of India.
(ii) It is not a part of the domestic income of India because the American embassy in India is considered as the domestic territory of U.S.A.
(iii) It is also not a part of the domestic income of India because the branches of the company are not located in the domestic territory of India.
(iv) It is considered as a part of the domestic income of (by non-residents) India since Indian embassy in China is considered to be the domestic territory of India.

9. How are the following treated in estimating the national income?
(i) Consultation fee by doctor.
(ii) Purchase of medicine by a patient.
(iii) Free uniform supplied to nurses in a hospital.
Ans. (i) Consultation fee received by a doctor is included in the national income since the doctor produces services for the patients.
(ii) Purchase of medicines by a patient is also included in the national income as it is the expenditure on purchase of medicines produced in an economy.
(iii) Free uniform supplied to nurses in a hospital is a part of total expenditure in an economy on goods and services. Therefore, it will be included in the national income.

10. What is gross value added of a firm?
Ans. Gross value added (GVA) of a firm Value of gross output-Value of intermediate consumption.

11. What is the net value added (NVA) of a firm?
Ans. NVA of a firm = Gross value added-Consumption of fixed capital.

12. How can you derive GNP_{mp} from GDP_{mp}?
Ans. GNP_{mp}=GDP_{mp}+ Net factor income earned abroad.

13. What is GDP deflator?
Ans. $GDP\ deflator = \frac{Nominal\ GDP\ in\ any\ year}{Real\ GDP\ in\ that\ year} \times 100$
14. Define corporate tax. How does taxation affect the level of circular flow?
Ans. It is a tax paid by an entrepreneur to the Government. It is a leakage in the circular flow. It reduces the flow.

15. Why are exports included in the estimation of domestic product by the expenditure method? Can gross domestic product be greater than gross national product? Explain.
Ans. Expenditure method estimates expenditure on domestic product, i.e., expenditure on final goods and services produced within the economic territory of the country. It includes expenditure by residents and non-residents both. Exports, though purchased by non-residents, are produced within the economic territory, and therefore, a part of domestic product. Domestic product can be greater than national product if factor income paid to the rest of the world is greater than the factor income received from the rest of the world i.e., when net-factor income received from abroad is negative.

16. Is net export a part of NFIA? Explain.
Ans. No, it is not. Net export, the difference between export and import (X-M), is a part of expenditure on domestic product. While NFIA is the difference between income earned from abroad by the normal residents of a country and income earned by non-residents in the domestic territory of that country. It is not included in the domestic product rather it is a component of NI. Therefore, both are different concepts.

17. When is the national income less than domestic factor income?
Ans. When NFIA is negative.

18. When is the national income larger than domestic factor income?
Ans. When NFIA is positive.

IMPORTANT FORMULAE AT A GLANCE

• Net indirect taxes	=	Market price - Factor cost
• Depreciation	=	Gross value - Net value
• Net factor income from abroad	=	National value - Domestic value
• GDP_{fc}	=	GDP_{mp} - Net indirect taxes
• NDP_{mp}	=	GDP_{mp} - Depreciation
• NDP_{fc}	=	GDP_{mp}-Net indirect taxes -Depreciation
• GNP_{mp}	=	GDP_{mp}+ Net factor income from abroad
• GNP_{fc}	=	GNP_{mp} - Net indirect taxes
. NNP_{mp}	=	GNP_{mp}- Depreciation
• NNP_{fc}	=	GNP_{mp}-Net indirect taxes - Depreciation

- **Private Income** = Income from domestic product accruing to private sector + Net factor income from abroad +Current transfers from Government + National debt interest + Net current transfers from the rest of the world.
- **Personal Income** = Private Income - Corporate tax - Retained earnings.
- **Personal Disposable Income** = Personal income - Personal taxes - Miscellaneous receipts of Government.

  Or

  = Personal consumption expenditure + Personal savings.
- **National Disposable Income** = National income + Net indirect taxes + Net current transfers from the rest of the world.
- **Gross National Disposable Income** = Net National disposable income + Depreciation.

Value Added Method

- **National Income** = Gross value added by primary sector + Gross value added by secondary sector + Gross value added by tertiary sector - Depreciation - Net indirect tax + Net factor income from abroad.
- **Value Added** = Value of output - Intermediate consumption
- **Value of Output** = Sales + Change in stock
- **Change in Stock** = Closing stock - Opening stock

Income Method

- **National Income** = NDPfc + Net factor income from abroad
- **NDP_{fc}** = Compensation of employees + Rent and royalty + Interest + Profit + Mixed income
- **NDP_{fc}** = Compensation of employees + Operating surplus + Mixed income
- **Compensation of Employees** = Wages and salaries in cash + Wages and salaries in kind + Employer's contribution to social security schemes
- **Operating Surplus** = Rent + Royalty + Interest + Profit
- **Profit** = Dividend + Undistributed profit + Corporate tax

Expenditure Method

- **National Income** = GDP_{mp} - Depreciation - Net indirect tax + Net factor income from abroad
- **GDP_{mp}** = Private final consumption expenditure + Government final consumption expenditure + Gross domestic capital formation + Net exports
- **Gross Domestic Capital Formation** = Gross fixed capital formation + Change in stock
- **Net Exports** = Exports - Imports.

NUMERICALS (SOLVED)

1. Calculate the value added by firm A and firm B from the following data: ₹ (in lakhs)

(i)	Purchase by firm A from the rest of the world	40
(ii)	Sales by firm B	100
(iii)	Purchases by firm A from firm B	60
(iv)	Sales by firm A	120
(v)	Exports by firm A	40
(vi)	Opening stock of firm A	45
(vii)	Closing stock of firm A	30
(viii)	Opening stock of firm B	40
(ix)	Closing stock of firm B	30
(x)	Purchases by firm B from firm A	60

Solution. Value added by firm A= (iv)+ [(vii)-(vi)](i)-(iii)=120+ [30-45]-40-60 = 5 lakhs.

Value added by firm B = (ii) + ((ix)-(viii)](x)=100+ [30-40]-60 = ₹ 30 lakhs.

2. Calculate N.I. by income and expenditure method: (in crores)

(i)	Subsidies	5
(ii)	Private final consumption expenditure	100
(iii)	NFIA	(-)10
(iv)	Indirect tax	25
(v)	Rent	5
(vi)	Government final consumption expenditure	20
(vii)	Net domestic fixed capital formation	30
(viii)	Operating surplus	20
(ix)	Wages	50
(x)	Net export	(-) 5
(xi)	Addition to stock	(-) 5
(xii)	Social security contribution by employers	10
(xiii)	Mixed income	40

Solution. Income method:

N.I.= (ix)+(xii)+(viii)+(xiii) (iii)=50+ 10+20+40+(-10) = 110 crores.

Expenditure method:

N.I.(ii)+(vi)+(vii) + (xi) + (x) − (iv) + (i) + (iii) 100+20+30+(-)5+(-)5-25+5+(-10)= 110 crores.

3.Estimate the following with the help of given data:,

(i)	GDP_{mp} (ii) Net Value Added at factor cost; and (iii) prove that it is equal to the income generated	₹(in crores)
(a)	Increase in the stock of unsold goods	1,000

(b)	Sales	10,000
(c)	Net indirect tax	800
(d)	Purchase of raw materials from other firms	1,650
(e)	Purchase of fuel and power	850
(f)	Consumption of fixed capital	500
(g)	Rent	700
(h)	Wages and salaries	3,500
(i)	Interest payment	1,000
(j)	Dividend	1,500
(k)	Corporate gain tax	300
(l)	Undistributed profit	200

Solution. (i) GDPmp = Sales + Increase in the stock - Purchase of raw materials - Purchase of fuel and power.
= 10,000+1,000-1,650-850-11,000-2,500 = 8,500 crores.

(ii) Net Value Added at factor cost = Sales + Increase in the stock - Purchase of raw materials - Purchase of fuel and power -Consumption of fixed capital - Net indirect tax.
= 10,000+1,000-1,650-850-500-800
= 11,000-3,800 = 7,200 crores.

(iii) Income generated = Rent + Wages and salaries + Interest + Dividend + Corporate gain tax + Undistributed profit.
=700+3,500+ 1,000+ 1,500+ 300+200 =₹ 7,200 crores.

Hence, it is proved that net value added at factor cost = Income generated.

4. Calculate:(i) gross domestic product at factor cost and (ii) net national disposable income. (in crores)

(a)	Net indirect tax	130
(b)	Government final consumption expenditure	100
(c)	Profit	90
(d)	Net Domestic Capital Formation	120
(e)	Change in Stocks	(-) 10
(f)	Private Final consumption expenditure	500
(g)	Net imports	20
(h)	Net current transfers to abroad	10
(i)	Net factor income to abroad	30
(j)	Gross domestic capital formation	160

Solution. (i)GDP_{fc} :
(f)+(b)-(g)+(j)-(a)
=500+100-20+160-130 = 760-150-610.

(i) NNDI: GDP_{fc} - Depreciation + NF IA + NIT - NET CURRENT TRANSFERS TO ABROAD
=610-(40)-30+130+(-10)-10=650.

5. From the following data calculate GNP at FC by:

(i) Income method, (ii) Expenditure method:	₹ (in crores)
(a) Net domestic capital formation	500
(b) Compensation of employees	1,850
(c) Consumption of fixed capital	100
(d) Govt. final consumption expenditure	1,100
(e) PVf. final consumption expenditure	2,600
(f) Rent	400
(g) Dividend	200
(h) Interest	500
(i) Net exports	(-) 100
(j) Profits	1,100
(k) NFIA	(-) 50
(l) Net indirect taxes	250

Solution. GNP_{fc}

(i) Income method:

=(b)+(f)+(h)+(i)+(k)

NNP_{fc} = 1,850+400+500 +1,100+(-50) = 3,800

GNP_{fc} = 3,800+100 = ₹3,900 crores.

(ii) Expenditure method = (a) + (c) + (d) + (e) + (i)+(k)-(l)

=500+100+1,100+2,600+(-100)+(-50)-250=₹3,900 crores.

6. There are only two producing sectors A and B in an economy. Calculate:

	₹ (in crores)
(a) Net factor income from abroad.	20
(h) Sales by A	1,000
(c) Sales by B	2,000
(d) Change in stock of B	(-)200
(e) Closing stock of A	50
(f) Opening stock of A	100
(g) Consumption of fixed capital by A and B	180
(h) Indirect taxes paid by A and B	120
(i) Purchase of raw material by A	500
(j) Purchase of raw material by B	600
(k) Exports by B	70

Solution. (i) GVA_{mp} of sector A= 1,000-50-500=450

GVA_{mp} of sector B = 2,000-200-600 = 1,200

Total = 450+ 1,200 = ₹1,650 crores.

(ii) NNPfc = 1,650-150-120+20 = ₹ 1,370 crores.

7. Calculate National income by the: (i) Expenditure method and (ii) Production method from the following data:

		₹ (in crores)
(a)	Gross value added at market price by the primary sector	300
(b)	PVT. final consumption expenditure	750
(c)	Consumption of fixed capital	150
(d)	Net indirect taxes	120
(e)	Gross value added at market price by the secondary sector	200
(f)	Net domestic fixed capital formation	200
(g)	Change in stock	(-)20
(h)	Gross value added at market price by the tertiary sector	700
(i)	Net imports	50
(j)	Government final consumption expenditure	150
(k)	Net factor income from abroad	-20

Solution. (i) N.I. by expenditure method = (b) + (j)+(f)+(g)(i)+(k)-(d)

=750+150+220+(-20)-50+20-120 = ₹ 950 crores.

(ii) N.I. by production method = (a) + (e) + (h) - (c) - (d) + (k) = 300+ 200+ 700-150-120+20= ₹ 950 crores.

8. From the following data show that net value added at factor cost (NVA) is equal to the sum of factor incomes.

		₹(in crores)
(i)	Purchase of raw material and other input from the domestic market	600
(ii)	Increase in stock	200
(iii)	Domestic sales	1,800
(iv)	Import of raw material	100
(v)	Exports	200
(vi)	Depreciation of fixed capital	75
(vii)	Salaries and wages	600
(viii)	Interest payments	450
(ix)	Rent	75
(x)	Dividents	150
(xi)	Undistributed profits.	80
(xii)	Corporate profit tax	20
(xiii)	Indirect tax	50

Solution. NVA_{fc} = (ii) + (iii) + (v) - (i) – (iv) - (vi)-(xiii)=200+ 1,800+200-600-100-75-50 = ₹ 1,375 crores.

Sum of factor income = (vii) + (viii) + (ix) + (x)+(xi) + (xii) = 600+450+75+150+80+20= ₹ 1,375 crores.

Hence, NVA_{fc} = Sum of factor income.

9. Calculate compensation of employees from the following data.

		₹(in crores)
(i)	Wages and salaries	2350
(ii)	Employer's contribution to social security system	40
(iii)	Value of free medical services	60
(iv)	Bonus	35

(v)	Employees contribution to social security scheme	30
(vi)	Value of rent free house	20
(vii)	Travelling allowance	15

Solution. Compensation of employees = (i) + (ii) + (iii) + (iv) + (vi) = 2505 crores.

10. Calculate operating surplus from the following data:

Items		₹(in lakhs)
(i)	Rent	400
(ii)	Interest	200
(iii)	Profit	300
(iv)	Dividends	50
(v)	Undristibuted profit	20

Solution. Operating surplus = Rent + Interest + Profit= ₹ 900 lakhs.

NUMERICALS (UNSOLVED)

1. From the information given below calculate:

	(i) Gross domestic product at market prices, (ii) Net national income at factor cost.	₹ (in crores)
(a)	Net domestic product at market price	27,500
(b)	Net indirect taxes	185
(c)	Income from domestic product accruing to Government	2,550
(d)	Net factor income from abroad	(-)230
(e)	Depreciation	1,200

[**Ans.** (i) ₹ 28700 crores, (ii) ₹ 27,085 crores]

2. From the data given below estimate: (i) Gross domestic product at market price, (ii) NNPfc

		₹ (in crores)
(a)	Gross domestic product at factor cost	3,200
(b)	Net factor income from abroad	(-) 100
(c)	Net indirect taxes	900
(d)	Consumption of fixed capital	250

[**Ans.** (i) ₹ 4,100 crores, (ii) ₹ 2,850 crores]

3. Calculate: (i) PDI, (ii) Personal income, (iii) Private income, from the following data.

Items		₹ (in crores)
(a)	Direct tax paid by the households	6,500
(b)	Corporation tax	1,500
(c)	Household final consumption expenditure	24,500
(d)	Interest on national debt	2,000
(e)	Saving of private corporate sector	3,500
(f)	Household savings	7,500

[**Ans.** (i) ₹ 3,200 crores, (ii) ₹ 38,500 crorers, (iii) ₹ 43500 crores]

4. Calculate: (i) GDP_{mp}. (ii) Private income and (iii) Personal income from the following data.

		₹ (in crores)
(a)	National income	32,500
(b)	Indirect taxes	3,350
(c)	Subsidies	850
(d)	Consumption of fixed capital	4,250
(e)	Net factor income from abroad	330
(f)	Income from entrepreneurship and property accruing to Government administrative departments	950
(g)	Saving of non-departmental enterprises	140
(h)	National debt interest	125
(i)	Current transfers from Government administrative Dept.	1,140
(j)	Other current transfers from rest of the world (net)	660
(k)	Saving of private corporate sector	320
(I)	Corporate profit tax	180

[**Ans.** (i)₹38,920 crores, (ii) ₹ 33,335 crores, (iii) ₹ 32,835 crores]

5. Calculate national income from the following data:

		₹ (in crores)
(i)	Personal income	2,450
(ii)	Saving of private corporate sector	25
(iii)	Corporation tax	45
(iv)	Current transfers from Government administrative departments	60
(v)	Current transfers from rest of the world	50
(vi)	Income from property and entrepreneurship accruing to the Government admn. Depts.	50
(vii)	Saving of non-department Government enterprises	40
(viii)	Net indirect taxes	390
(ix)	Direct taxes paid by households	50

[**Ans.** ₹ 2,500 crores]

6. From the following data about a firm X, calculate net value added at factor cost:

		₹ (in crores)
(i)	Sales	350
(ii)	Opening stock	30
(iii)	Closing stock	20
(iv)	Purchases of machinery	150
(v)	Purchases of intermediate products	170
(vi)	Subsidy	40
(vii)	Depreciation	35

[**Ans.** ₹ 175 thousand]

7. From the following data about a firm estimate the net value added at factor cost by the firm:

		₹ (in crores)
(i)	Domestic sales	2,000
(ii)	Subsidies	40
(iii)	Purchase of raw material	560
(iv)	Exports	500

(v)	Depreciation	100
(vi)	Import of raw materials	60
(vii)	Change in stock	200
(viii)	Net indirect taxes	60

8. From the following data, calculate gross value added at factor cost: (in lakhs)

(i)	Sales	180
(ii)	Rent	5
(iii)	Subsidies	10
(iv)	Change in stock	15
(v)	Purchase of raw materials	100
(vi)	Profits	25

[**Ans.** ₹ 105 lakhs]

9. Calculate gross value added at factor cost from the following data:

(i)	Sales tax	20
(ii)	Sales	400
(iii)	Purchase of raw material	250
(iv)	Excise duty	30
(v)	Change in stocks	- 40
(vi)	Import of raw material	12
(vii)	Depreciation	9

[**Ans.** ₹ 60 lakhs]

10. Calculate net value added at factor cost from the following data: ₹ (in crores)

(i)	Sales	2,200
(ii)	Used for self consumption by the owners	300
(iii)	Opening stock	350
(iv)	Purchases of raw materials	1,050
(v)	Electricity charges	80
(vi)	Consumption of fixed capital	130
(vii)	Excise duty	60
(viii)	Income tax	30
(ix)	Closing stock	100

[**Ans.** ₹ 930 crores]

11. Calculate gross value added at factor cost: ₹ (in crores)

(i)	Unit of output added at factor cost:	1,000
(ii)	Price per unit output	30
(iii)	Depreciation	1,000
(iv)	Intermediate cost	12,000

(v) Closing stock	3,000
(vi) Opening stock	2,000
(vii) Excise	2,500
(viii) Sales tax	3,500

[**Ans.** ₹ 13,000 crores]

12. Find gross value added at factor cost:

	₹ (in crores)
(i) Unit of output sold (Units)	2,000
(ii) Price per unit output	20
(iii) Depreciation	2000
(iv) Change in stock	(-)500
(v) Intermediate cost	15000
(vi) Subsidy	3,000

[**Ans.** ₹27,500 crores]

13. An economy has only two firms A and B. On the basis of the following information about these firms, find out:

(i) Value added by firm A and B	1000
(ii) Gross domestic product at market price.	30
	₹ (in lakhs)
(a) Export by firm A	30
(b) Import by firm A	60
(c) Sales to households by firm A	100
(d) Sales to firm B by firm A	50
(e) Sales to firm A by firm B	40
(f) Sales to households by firm B	80

[**Ans.** (i) Value added by firm A = ₹ 80 lakhs, By firm B = ₹ 70 lakhs, (ii) Gross domestic product at market price = ₹ 150 lakhs]

14. Calculate the value added by firm A and firm B with the help of the following data:

	(in lakhs)
(i) Purchase by firm A from the rest of the world	30
(ii) Sales by firm B	90
(iii) Purchase by firm A from firm B	50
(iv) Sales by firm A	110
(v) Exports by firm A	30
(vi) Opening stock of firm A	35
(vii) Closing stock of firm A	20
(viii) Opening stock of firm B	30
(ix) Closing stock of firm B	20
(x) Purchase by firm B from firm A	50

[**Ans.** Value added by firm A= ₹ 15 lakhs; Value added by firm B = ₹ 30 lakhs]

Hint: Item No. (v) is a part of item No. (iv)

15. On the basis of following data, calculate: (i)Gross value added market price for primary, secondary and tertiary sectors and (ii)National income. ₹ (in crores)

(a)Value of output of:

(I)	Primary sector	1,800
(II)	Secondary sector	1,200
(III)	Tertiary sector	1,300

(b) Value of intermediate inputs purchased by:

(I)	Primary sector	800
(II)	Secondary sector	400
(III)	Tertiary sector	350
(c)	Indirect taxes paid all the sectors	150
(d)	Consumption of fixed capital of all sectors	180
(e)	Factor income received by residents from rest of world	90
(f)	Factor income paid to non- residents	120
(g)	Subsidies received by all the sectors	80

[**Ans.** (i) Primary sector ₹ 1,000 crores: Secondary sector = ₹ 800 crores; Tertiary sector ₹ 950 crores, (ii) ₹ 2,470 crores]

16. From the following data calculate national income by income and expenditure method: (in crores)

(i)	Government final consumption expenditure	5,000
(ii)	Indirect taxes	6,000
(iii)	Subsidies	2,200
(iv)	Mixed income of self employed	17,100
(v)	Gross fixed capital formation	10,000
(vi)	Net addition to stocks	2,000
(vii)	Operating surplus	8,000
(viii)	Consumption of fixed capital	2,500
(ix)	Private final consumption expenditure	31,000
(x)	Exports of goods and services	4,600
(xi)	Imports of goods and services	5,200
(xii)	Net factor income from abroad	(-) 100
(xiii)	Compensation to employee	16,000

[**Ans.** ₹ 41,000 crores]

17. Calculate GNP_{fc} by: (i) income method and (ii) expenditure method from the following data. (in lakhs)

(a)	Rent	40
(b)	Private final consumption expenditure	800
(c)	Net exports	20
(d)	Interest	60
(e)	Profit	120
(f)	Government final consumption expenditure	200
(g)	Net domestic capital formation	100
(h)	Compensation of employees	800
(i)	Consumption of fixed capital	20

(j)	Net indirect taxes	100
(k)	Net factor income from abroad	(-) 20

[**Ans.** (i) Income method - 1,020 lakhs, (ii) Expenditure method - 1,020 lakhs]

18. From the following data, calculate gross National product at market price by: (i) income method and (ii) expenditure method.

		₹ (in crores)
(a)	Net domestic capital formation	375
(b)	Compensation of employees	600
(c)	Net indirect taxes	150
(d)	Profits	450
(e)	Rent	200
(f)	Private final consumption expenditure	1,100
(g)	Consumption of fixed capital	115
(h)	Government final consumption expenditure	700
(i)	Interest	250
(j)	Mixed income of self-employed	200
(k)	Net factor income from abroad	(-) 15
(l)	Net exports	(-) 25

[**Ans.** (i) ₹ 2,250 crores, (ii) ₹ 2,250 crores]

19. Find out: (i) net domestic product at factor cost by expenditure method and (ii) gross domestic product at market price by income method from the following data.

		₹ (in crores)
(a)	Government final consumption expenditure	150
(b)	Private final consumption expenditure	1,020
(c)	Compensation of employees	490
(d)	Rent, interest and profits	190
(e)	Export	100
(f)	Imports	110
(g)	Indirect taxes	180
(h)	Net fixed capital formation	180
(i)	Net factor income from abroad	(-) 5
(j)	Mixed income of the self employed	560
(k)	Government current transfers to households	20
(l)	Change in stocks	60
(m)	Consumption of fixed capital	80
(n)	Subsidies	20

[**Ans.** (i) ₹ 1,240 crores, (ii) ₹ 1,480 crores]

20. From the following data, calculate: (i) national income and (ii) personal disposable income.

		₹ (in crores)
(a)	Government final consumption expenditure	1,500
(b)	Change in stock	60
(c)	Gross domestic capital formation	800
(d)	Private income	4,000
(e)	Net exports	(-)70

(f)	Corporation tax	500
(g)	Net indirect taxes.	250
(h)	Private final consumption expenditure	2,800
(i)	Net factor income from abroad	(-)50
(i)	Consumption of fixed capital	200
(k)	Net retained earnings of private enterprises	50
(l)	Direct taxes paid by households	300

[**Ans.** (i) ₹ 4,530 crores, (ii) ₹ 3,150 crores]

21. Calculate: (i) Net domestic product at factor cost and (ii) Net national disposable income from the following data:

Items		(in lakhs)
(a)	Net indirect tax	60
(b)	Wages and salaries	600
(c)	Net current transfers from abroad	(-) 10
(d)	Rent	50
(e)	Interest	250
(f)	Profit	450
(g)	Net factor income from abroad	(-) 20
(h)	Royalty	40
(i)	Social security contributions by employers	100

[**Ans.** (i) ₹ 1,140 lakhs, (ii) ₹ 1,170 lakhs]

22. From the following data calculate: (i)NDP_{mp} and (ii)NNP_{fc} by income and expenditure methods.

		₹ (in crores)
(a)	Personal consumption expenditure	610
(b)	Wages and salaries	230
(c)	Employers contribution to social security schemes	200
(d)	Gross business fixed investment	180
(e)	Profits	50
(f)	Gross residential construction investment	120
(g)	Government purchases of goods and services	95
(h)	Gross public investment	60
(i)	Rent	70
(j)	Inventory investment	55
(k	Exports	50
(l)	Interests	60
(m)	Imports	60
(n)	Net factor income from abroad	(-) 5
(o)	Mixed income	380
(p)	Depreciation	40
(q)	Subsidies	10
(r)	Indirect taxes	90

[**Ans.** (i) NDP_{mp} = ₹ 1,070 crores, (ii) NNP_{fc} = 985 crores]

23. Calculate: (i)national income and (ii)compensation of emoployees from the following data.

		₹ (in crores)
(a)	Royalty	15
(b)	Private final consumption expenditure	400
(c)	Net current transfers from rest of the world	20
(d)	Government final consumption expenditure	100
(e)	Net factor income from abroad	(-) 10
(f)	Net domestic capital formation	80
(g)	Consumption of fixed capital	50
(h)	Net exports	40
(i)	Net indirect taxes	60
(j)	Employer's contribution to social security schemes	90
(k)	Rent	180
(l)	Interest	120
(m)	Profits	85

[**Ans.** (i) ₹ 550 crores, (ii) ₹ 160 crores]

24. From the following data, calculate NDP_{fc}:

		₹ (in crores)
(i)	Change in stocks	20
(ii)	Government final consumption expenditure	100
(iii)	Private final consumption expenditure in domestic market	500
(iv)	Net exports	(-) 20
(v)	Direct purchases from abroad by resident households	50
(vi)	Net factor income from abroad	10
(vii)	Net indirect taxes	60
(viii)	Direct purchases by non-resident in domestic market	150
(ix)	Gross domestic capital formation	100
(x)	Consumption of fixed capital	20

[**Ans.** NDP_{fc} = ₹ 500 crores]

25. From the following data calculate: national income by income method.

		₹ (in crores)
(i)	Government final consumption expenditure	100
(ii)	Subsidies	10
(iii)	Rent	200
(iv)	Wages and salaries	600
(v)	Indirect taxes	60
(vi)	Private final consumption expenditure	800
(vii)	Gross domestic capital formation	120
(viii)	Social security contribution by employers	55
(ix)	Royalty	25
(x)	Net factor income paid to abroad	30
(xi)	Interest	20
(xii)	Consumption fixed capital	10
(xiii)	Profit	130
(xiv)	Net exports	70
(xv)	Change in stock	50

[**Ans.** N.I. by income method = ₹ 1,000 crores]

WORKSHEET FOR SELF EVALUATION

1. What is national product? **(2 Marks)**
2. Which of the national income aggregates is technically known as national income? **(2 Marks)**
3. Give two examples of transfer payments. **(2 Marks)**
4. What should be deducted from gross domestic to get the net domestic product? **(2 Marks)**
5. Define GNP at factor cost. How is it different from national income? **(3 Marks)**
6. Can the net factor income from abroad be negative? **(2 Marks)**
7. What is economic welfare? **(2 Marks)**
8. What is meant by double counting? **(2 Marks)**
9. State the equilibrium condition of circular flow in two-sector economy. **(3 Marks)**
10. Distinguish between real flows and money flows. **(4 Marks)**
11. Differentiate between closed economy and open economy. **(4 Marks)**
12. What do you mean by leakages? Give two examples of it. **(4 Marks)**
13. Distinguish between transfer payments and factor payments. How are they treated in the measurement of national income? **(2 Marks)**
14. Define mixed income. **(2 Marks)**
15. State the main components of domestic factor income. **(3 Marks)**